The Missing Mergers: Modelling the observed binary neutron-star population and their mergers

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Abstract

The dawn of the age of gravitational wave astronomy heralds exciting results and new challenges. Chief among these is an accurate theory of binary mergers. When two massive stars in a binary survive their two supernovae, and produce a pair of Neutron Stars, they can merge, producing a Gravitational Wave transient event. Critically, then, providing a consistent estimate of the merger rates of compact binaries, as well as accurately predicting the period-eccentricity distributions of such binaries, is of utmost importance. We investigate the mechanism for natal supernova kicks, and in particular attempt to refine the Bray kick, which is governed by the equation $v_{\rm kick} = \alpha Q + \beta$, where Q is the ejecta-remnant mass ratio. We find that the kick has a characteristic slope α and offset β of either $\alpha = 110 \pm 5$ km/s, $\beta = 0$ km/s or $\alpha = 40 \pm 5$ km/s, $\beta = 100 \pm 5$ km/s. These parameters are sufficient to replicate the LIGO O3 data results. We find that $\alpha = 110 \pm 5$ km/s, $\beta = 0$ km/s has a merger rate of $\log\left(\mathcal{R}_0 \ / \ \mathrm{yr}^{-1}\ \mathrm{Gpc}^{-3}\right) = 2.5140 \pm 0.0002$, and $\alpha = 40 \pm 5$ km/s, $\beta = 100 \pm 5$ km/s has a merger rate of $\log\left(\mathcal{R}_0 \ / \ \mathrm{yr}^{-1}\ \mathrm{Gpc}^{-3}\right) = 2.526 \pm 0.009$. Moreover we find that the $\beta = 0$ model has the lowest frequency of highly eccentric extant NSNS systems out of all models considered, however this frequency exceeds our conservative requirements. We present a consistent and intuitive method for computing period-eccentricity distributions, and compute these for the models considered. We use this to illustrate the importance of predicting certain pulsars such as J1930-1852 (Swiggum et al., 2015). Consequently, we conclude that if the Bray kick is correct, it must be incomplete, and conjecture a new model for the Bray kick for future research to consider.

We all change, when you think about it. We are all different people all through our lives and that's okay, that's good, you've got to keep moving. So long as you remember all the people that you used to be. I will not forget one line of this, not one day, I swear. I will always remember when The Doctor was me.

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Everyone you love, everyone you know, everyone you ever heard of, every human being who ever was, lived out their lives on a mote of dust suspended in a sunbeam.

Carl Sagan

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Table of Notation

Unless context otherwise requires, the following symbols and variables have the following meanings in this thesis. A superscript asterisk* denotes an assumed quantity.

Symbol	Meaning	Numerical Value	Units
z	Cosmological redshift	-	Dimensionless
Z	Metallicity	-	(mass fraction)
ψ	Stellar formation rate density (SFRD)	-	$M_\odot$ / yr / Mpc^3
χ	Stellar formation history (SFH)	-	$M_\odot$ / Mpc^3
H_0	Hubble constant	69.6	km / s / Mpc
Ω_M	Matter density of Universe	$\frac{8\pi G\rho_0}{3H_0^2} = 0.3^*$	Dimensionless
Ω_Λ	Cosmological constant density of Universe	$\frac{\Lambda c^2}{3H_0^2} = 0.7^*$	Dimensionless
Ω_k	Curvature of Space Parameter	$1 - \Omega_\mathrm{M} - \Omega_\Lambda$	Dimensionless
$Z_\odot$	Solar Metellacity	0.020^*	(mass fraction)

Furthermore, a subscript zero (e.g., H_0) in a cosmological parameter represents "as at the current epoch", whilst a subscript $\odot$ represents a quantity measured in relation to the sun (e.g., $r_\odot$ is solar radius, $M_\odot$ is solar mass)

On tensors, Latin indices $(i, j, k \dots)$ run from 1 to 3, whereas Greek indices $(\alpha, \beta, \delta \dots)$ run from 0 to 3. For our tensor calculations, we use the $(-+++)$ (or positive trace, or $\mathrm{Tr}\{\eta\} = +2$) metric.

Glossary of Terms

Notation	Description
BHBH	Black Hole-Black Hole.
BPASS	Binary Population and Spectral Synthesis code.
CCSNe	Core-Collapse Supernovae.
CEE	Common Envelope Evolution.
DTD	Delay-Time Distribution.
IMF	Initial Mass Function.
KAGRA	Kamioka Gravitational Wave Detector.
KDE	Kernel Density Estimation.
LIGO	Laser Interferometer Gravitational-Wave Observatory.
LISA	Laser Interferometer Space Antenna.
NSBH	Neutron Star-Black Hole.
NSNS	Neutron Star-Neutron Star.
NSWD	Neutron Star-White Dwarf.
PISN	Pair Instability Supernovae.
SFR	Stellar Formation Rate.

Notation	Description
SFRD	Stellar Formation Rate Density.

Chapter 1

Introduction

It has been said that astronomy is a humbling and character-building experience. There is perhaps no better demonstration of the folly of human conceits than this distant image of our tiny world. To me, it underscores our responsibility to deal more kindly with one another and to preserve and cherish the pale blue dot, the only home we've ever known.

Carl Sagan

It is no secret that humanity has always harboured a deep and burning passion for the heavens. As children, we make wishes on shooting stars. We gaze upon them in wondrous excitement, reminded of the vast history of the Universe and the unimaginable future of it. We even turn to them for guidance – literal guidance, as my ancestors, the early Polynesian sailors, used them to navigate the South Pacific – and to wonder. Staring into the stars is, in some sense, the very essence of humanity, and it often provokes a flurry of anthropocentric questions.

Writing an introduction to Physics for new students at Princeton, Albert Michelson once wrote (Michelson, 1896)

> ...it seems probable that most of the grand underlying principles have been firmly established ... An eminent physicist remarked that the future truths of physical science are to be looked for in the sixth place of decimals.

Two decades later, Albert Einstein had discovered General Relativity, and Quantum Mechanics was rapidly emerging as a new field of interest to most physicists. General Relativity in particular revolutionised modern Physics: from the perihelion precession of Mercury, to the microlensing of starlight around massive bodies, it brought about resolutions to age old conundrums, and piqued the interest of a new generation of astronomers. However some of General Relativity's most thrilling predictions – entering even the realm of popular science – were the existence of black holes and Gravitational Waves.

But to understand Gravitational Waves, one must understand and appreciate stellar evolution. When massive stars ($8M_\odot \lesssim M \lesssim 80M_\odot$) become sufficiently old, they violently explode in supernovae, as postulated by Baade and Zwicky (1934); Hoyle (1946). In a binary star system, which this thesis concerns itself with, this occurs twice – once for the primary (more massive) star, and once for the secondary (less massive) star. When these two supernovae have occurred, two compact remnants remain: black holes and neutron stars.

In 1916, Einstein showed that in the weak-field limit of the Einstein Field Equations, one sees a wave equation in space-time emerge (Einstein, 1916), all but confirming Poincaré's hypothesis that such waves exist. These were theorised to originate from the inspiral of a pair of compact objects, as well as through events like Core-Collapse Supernovae (CCSNe) and Pair Instability Supernovae (PISN). But for the next near-century, these waves remained invisible to researchers; no successful detections were made until 2015, when the Laser Interferometer Gravitational-Wave Observatory (LIGO) detected (Abbott et al., 2016a) the presence of Gravitational Waves, by using two interferometers located at LIGO Livingston (in Louisiana) and LIGO Hanford (in Washington State). Since then more than 50 Gravitational Wave events have been detected by the LIGO detectors, which exist in partnership with the VIRGO collaboration in Europe. With the commissioning of the Kamioka Gravitational Wave Detector (KAGRA) and eventual launch of the Laser Interferometer Space Antenna (LISA) interferometers, more events will be detected. In particular, KAGRA is estimated to observe between 5.4 and 8.2 Neutron Star-Neutron Star (NSNS) mergers per year, and between 0.07 and 7 Black Hole-Black Hole (BHBH) mergers per year (Kawamura et al., 2009) – both rates being dependent on the operating mode. LISA is expected to observe around 8.75 NSNS binaries per year (Lau et al., 2020) and around 15 BHBH binaries per year (Rhook and Wyithe, 2005).

Once the first detection of a pair of coalescing black holes (GW150914) had been made, the LIGO collaboration was immediately in a position to compute expected merger rates of the system; the estimate presented in Abbott et al. (2016a) constrained this binary black hole rate to be between 2-400 $\mathrm{Gpc}^{-3}\ \mathrm{yr}^{-1}$. More recent studies such as Abbott et al. (2020a) put it at $23.9^{+14.9}_{-8.6}\ \mathrm{Gpc}^{-3}\ \mathrm{yr}^{-1}$. This information – for the three classes of compact remnants we are concerned with – is presented in Table 1.1 on page 4.

The overarching theme of this thesis is to investigate the impact that varying the form of the NSNS kick has on two observable parameters:

(1) The merger rate of NSNS systems we detect today, and

(2) The theoretical period-eccentricity distributions of NSNS systems in our Galaxy.

For both of these parameters, we will investigate how modifying the form and magnitude of the kick results in theoretical period-eccentricity distributions that can replicate observed distributions.

We consider three classes of compact remnants – NSNS systems, Neutron Star-Black Hole (NSBH) systems, and BHBH systems. Although other classes of compact remnants exist, such as Neutron Star-White Dwarf (NSWD) binaries, they are beyond the scope of this thesis. This is due to LIGO at present only reporting merger rate densities for NSNS, NSBH, and BHBH systems.

1.1 | Gravitational Wave Astronomy

Gravitational Wave Astronomy was first established as a scientific discipline with the publication of Einstein (1916), in which Einstein derived a wave equation for spacetime. Since then, attempts at direct detection of Gravitational Waves have been advertised. However as the wave equation emerges only in the weak-field limit of the Einstein Field Equations, a considerable amount of noise reduction is required, and detection was not accomplished until 2015, when LIGO detected the merger of a binary black hole system, as outlined in Abbott et al. (2016a). This merger, GW150914, was detected on September 14, 2015, and consisted of two Black Holes of ~ 35 $M_\odot$ each. The signal lasted for < 0.5 seconds, peaking at a maximum strain of 10^{-21} (no units) (Abbott et al., 2016b).

One of the seminal publications in this field was Peters (1964). Peters' approach to mathematically modelling the decaying orbits of a binary system is often referred to as the "Peters Presciption" and provides exact analytical expressions for $\frac{da}{dt}$ and $\frac{de}{dt}$ (i.e. the coupled differential equations of motion of the orbit, representing the change in semimajor axis a and eccentricity e with time). These relations are further explored in Nyadzani and Razzaque (2020), and used to critically evaluate the coalescence time expression, numerically solve it, and provide analytical approximations (as the integral in the expression is not well-formed).

1.1.1 | Efforts to detect Gravitational Waves

Abbott et al. (2016a) represents the first direct detection of Gravitational Waves. The relative difficulty of detecting Gravitational Waves is no better exemplified than by Abadie et al. (2011), who show that a NSNS merger on the other side of the Galaxy – let alone outside of our Local Group – would produce a strain on the order of $h \simeq 10^{-19}$. This is four orders of magnitude smaller than the radius of a proton. Consequently, the dominant source of noise in any interferometer is quantum shot noise – that of a photon colliding with the reflector. Thus, it was not until the 21st century when detectors such as LIGO could be built to the relevant precision required to detect such events. Notwithstanding this increase in precision, LIGO must correct for seismic noise and acoustic noise (such as trucks driving near the detectors). This is mitigated by the fact the two detectors in the United States are constructed 3,000 km apart – noise in one interferometer will generally not be present in the other. Hence, signals that appear in *both* detectors at the same time will *generally* be true signals.

Moreover, until very recently, the generation of squeezed light was a limiting factor in the detection of Gravitational Waves. As discussed in Abadie et al. (2011) and first proposed in Caves (1981), the sensitivity of the Gravitational Wave detectors can be increased by using squeezed light. In summary, this is the process of decreasing the photon counting error at the expense of increasing the radiation-pressure error – analogous to the Heisenberg Uncertainty Principle.

1.1.2 | Merger Rate Densities

As will be explored in Chapter 2, the merger rate is characterised by the *merger rate density*, i.e., the number of mergers detected per unit time, per unit volume. We denote this by $\mathcal{R}_0$. Great effort (see Abbott et al. 2016a, 2019, 2020a) has been undertaken to compute $\mathcal{R}_0$, and the best estimates from the LIGO collaboration can be found in Table 1.1. It is important to note that the singular bound on the NSBH merger rate is due to the difficulties in obtaining a definitive detection candidate. Thus although we may refine the theory to match the observations, and potentially narrow the result significantly, there will be an intrinsic statistical error in the prediction for NSBH mergers. To mitigate this error (across all models considered), we can evaluate the resulting period-eccentricity distributions from evolving the modelled binaries.

	Empirical			Theoretical[‡] (BPASS)	
	$\frac{\mathcal{R}_0}{\mathrm{yr}^{-1}\ \mathrm{Gpc}^{-3}}$	$\log\left(\frac{\mathcal{R}_0}{\mathrm{yr}^{-1}\ \mathrm{Gpc}^{-3}}\right)$	Ref.	$\frac{\mathcal{R}_0}{\mathrm{yr}^{-1}\ \mathrm{Gpc}^{-3}}$	$\log\left(\frac{\mathcal{R}_0}{\mathrm{yr}^{-1}\ \mathrm{Gpc}^{-3}}\right)$
NSNS	320^{+490}_{-240}	$2.5^{+0.6}_{-0.4}$	*	355 ± 16	2.55 ± 0.02
NSBH	< 610	< 2.79	†	236 ± 16	2.37 ± 0.03
BHBH	$23.9^{+14.9}_{-8.6}$	$1.7^{+0.4}_{-0.1}$	*	162 ± 11	2.21 ± 0.03

Table 1.1: Merger Rate Densities (in $\mathrm{yr}^{-1}\ \mathrm{Gpc}^{-3}$) for different transient types as measured by the LIGO collaboration. Whilst bounds can be placed on the "twin remnant" systems, only a loose upper limit can be placed on the mixed NSBH model due to the lack of confident detection candidates (as discussed in Section 7c of Abbott et al. (2019)). It is also important to note that, in line with Eldridge et al. (2018), the theoretical event rate densities are shown as $\log \mathcal{R}_0$ values. This is represented again in Fig. 1.2.
Sources:
* Abbott et al. (2020a)
† Abbott et al. (2019)
‡ Eldridge et al. (2018)

1.1.3 | Period-Eccentricity distributions

A period-eccentricity distribution is a two-dimensional scatter plot of period vs. eccentricity. The period and eccentricity fully describe a binary system with constant masses and denser regions of a period-eccentricity plot correspond to more likely observations. We infer this plot observationally by plotting the periods and eccentricities of each NSNS system in the period-eccentricity plane. By comparison, we can generate a theoretical plot as a probability distribution, where each point in the plane corresponds to a probability of observing that

period and eccentricity pair. We can quantify the goodness-of-fit of the model (i.e., compute the "best model") by comparing the observational data to the theoretical. The best model will be the one that reproduces the observed distribution in Table 1.2 most closely. We also show this distribution in Fig. 1.1. We highlight the pulsar J1930-1852 as "its unique spin and orbital parameters challenge models that describe DNS formation" (Swiggum et al., 2015).

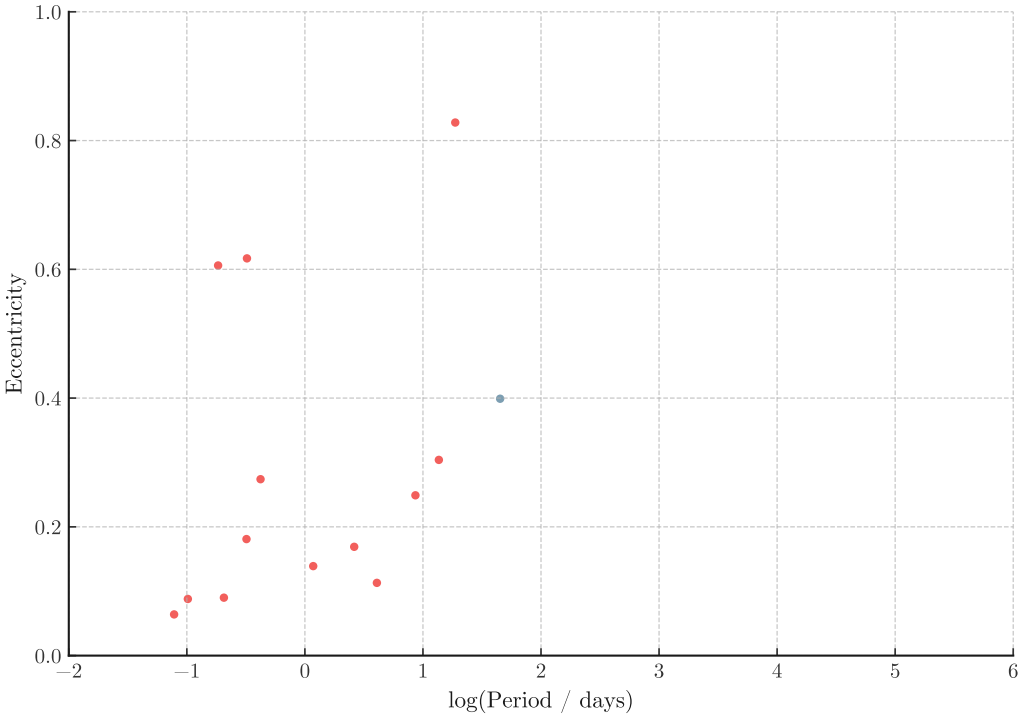


Figure 1.1: The period-eccentricity distribution of known NSNS systems. The systems are those shown in Table. 1.2. The blue point is the pulsar J1930-1852 from Swiggum et al. (2015).

Our theoretical Period-Eccentricity distributions are generated by creating a cube, with period, eccentricity, and time dimensions. A system with a given initial period and eccentricity will trace out a path in this space, and we can plot it as a single plot by projecting this 3D space into the period-eccentricity plane.

1.2 | Understanding the evolution of binary systems

The actual population synthesis for our analysis is performed by Tui, part of the BPASS suite of codes (Eldridge et al., 2017; Stanway and Eldridge, 2018). Here we present a brief overview of the key points of binary population synthesis, but refer the reader to the papers

Pulsar	log (period / days)	e
J0453+1559	0.609808	0.113
J0737-3039	-0.991400	0.088
B1534+12	-0.375718	0.274
J1756-2251	-0.494850	0.181
B1913+16	-0.490797	0.617
J1913-1102	-0.686133	0.090
J1757-1854	-0.735182	0.606
J1518+4904	0.936212	0.249
J1811-1736	1.273672	0.828
J1829+2456	0.070407	0.139
J1930-1852	1.653791	0.399
J1753-2240	1.134751	0.304
J1411+2551	0.417638	0.169
J1946+2052	-1.107905	0.064

Table 1.2: Selection of observed pulsar data, from Vigna-Gómez et al. (2018).

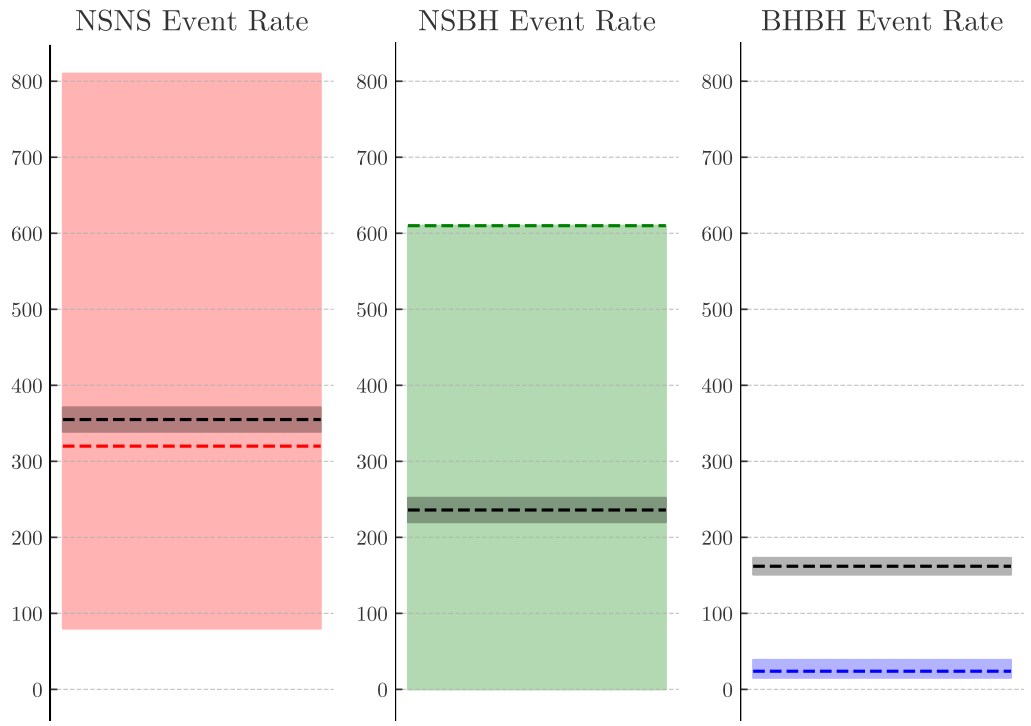


Figure 1.2: The Binary Population and Spectral Synthesis code (BPASS) vs. LIGO event rate density plots. The BPASS prediction is shown as a black dashed line in each plot, and the shaded region around each dashed line is the uncertainty associated with it. For exact values, see Table. 1.1.
Sources:
NSNS and BHBH event rate: Abbott et al. (2020a)
NSBH event rate: Abbott et al. (2019)
BPASS event rate: Eldridge et al. (2018)

for a more comprehensive discussion on the population synthesis. The two models we use are the Hobbs kick (Hobbs et al., 2005) and the Bray kick, (Bray, 2018) though other kick models exist (such as Baker et al. (2008)). Other population synthesis models include COMPAS (Barrett et al., 2016) and StarTrack (Belczynski et al., 2002), however we do not discuss them in detail here.

1.2.1 | Population Synthesis of NSNS systems

Stellar populations are defined by various parameters such as their masses, ages, eccentricities, semimajor axes, and metallicities. To generate theoretical datasets that we can work with, we must sample each of these parameters from their relevant probability densities or distribution functions.

The first parameter is mass; this is defined by the Initial Mass Function (IMF). We use the broken power-law IMF relation given in Stanway and Eldridge (2018); Kroupa et al. (1993). This has four main parameters: α_1, α_2 (dimensionless constants parameterising the gradient of the IMF – i.e., $\frac{dN}{dM} \propto M^{\alpha_i}$ over a low mass range for α_1 or high mass range for α_2), M_1 being the mass threshold at which the exponent changes, and $M_{\max}$ the maximum mass under consideration. We take $\alpha_1 = -1.30, \alpha_2 = -2.35, M_1 = 0.5M_\odot, M_{\max} = 300M_\odot$, giving

$$N(M < M_{\max}) \propto \int_{0.1}^{0.5} \left(\frac{M}{M_\odot}\right)^{-1.3} \mathrm{d}M + 0.5^{-1.3} \int_{0.5}^{M_{\max}} \left(\frac{M}{M_\odot}\right)^{-2.35} \mathrm{d}M \tag{1.1}$$

For stars that have a companion (the multiplicity of which is given by Moe and Di Stefano (2017) and repeated in Table 1.3), we take the mass ratio q to be given by a power-law distribution dependent on the period P of the binary, as in Moe and Di Stefano (2017). This prescription takes the form $p_q \propto q^\gamma$, where γ can be determined as a function of both q and $\log P$, and p_q is the probability of selecting the mass ratio q.

Explanation	early-B $9M_\odot \leq M_1 \leq 16M_\odot$	O-type $M_1 \geq 16M_\odot$
Single Stars	$16 \pm 9\%$	$6 \pm 6\%$
Binary Stars	$32 \pm 10\%$	$21 \pm 11\%$
Triple + Quadruple Stars	$52 \pm 13\%$	$73 \pm 16\%$

Table 1.3: The multiplicity of massive stars, from Moe and Di Stefano (2017). More massive stars have higher multiplicity – and the vast majority of massive stars have at least one companion.

Tui calculates the eccentricity e in the same way as Tauris and Takens (1998), namely it assumes the binary is circular after the first supernova, and then selects an eccentricity for the final orbit with the same semilatus rectum $a\left(1-e^2\right)$ as the circular orbit's radius. This computes the semimajor axis as a byproduct.

Finally, the metallicities are taken from a known sample, given in Table 1.4. This produces 13 sets of models, one corresponding to each metallicity. A single model is a space-delimited set of parameters. Table 1.5 gives the first five lines of the $Z = Z_{\odot}$ model output from Tui that we use as input to our code. This particular set of models was generated with the Bray kick. The seven parameters Tui generates are:

(1) Mass of the primary remnant (in solar masses),

(2) Mass of the secondary remnant (in solar masses),

(3) Initial semimajor axis of the resultant binary (in solar radii),

(4) Initial eccentricity of the binary,

(5) The number of binary systems we expect to see with these orbital parameters, per 10^6 solar masses of star formation,

(6) The rejuvenation age (the age at which the primary star last accreted mass) of the binary (in years), and

(7) The evolution age (how long the primary star spends on the main sequence) of the binary (in years).

1.3 | Thesis Outline

This thesis tours the evolution of compact binaries, beginning with a broad theoretical overview in Chapter 2. We proceed to a discussion on the methodology undertaken, and on the principles of population synthesis in Section 2.3, and discuss the predictions for event rate densities for the various kick models in Chapter 3. The generation of Period-Eccentricity distributions for NSNS systems is further discussed in Chapter 4, before we conclude in Chapter 5. In the Appendices, we variously derive Kepler's Laws (Appendix A), and provide various codes discussed throughout the thesis (Appendix B – E).

Z (mass fraction)	$\frac{Z}{Z_\odot}$ (no dim.)	BPASS name	Langer Prefactor (at $z = 0$, 4 s.f.)
10^{-5}	0.0005	zem5	4.510×10^{-5}
10^{-4}	0.005	zem4	0.002113
0.001	0.05	z001	0.01148
0.002	0.1	z002	0.01838
0.003	0.15	z003	0.02395
0.004	0.2	z004	0.04447
0.006	0.3	z006	0.07163
0.008	0.4	z008	0.08127
0.01	0.5	z010	0.1304
0.014	0.7	z014	0.2100
0.02	1.0	z020	0.2438
0.03	1.5	z030	0.1290
0.04	2.0	z040	0.03346

Table 1.4: The BPASS metallicites for population synthesis. The BPASS name field is an encoding of the metallicity for filename purposes within BPASS. Each metallicity contributes in a different amount to the stellar formation history of the Universe – see eqn(2.65) for details. The "Langer Prefactor" is the coefficient to $\psi(z)$ in eqn(2.65), representing the weight to assign to that Stellar Formation Rate (SFR).

$\frac{m_1}{M_\odot}$	$\frac{m_2}{M_\odot}$	$\frac{a_0}{R_\odot}$	e_0	weight	$\frac{t_{\rm rej}}{\rm yr}$	$\frac{t_{\rm evo}}{\rm yr}$
6.037700	1.468070	278.2180	0.6605925	0.3285020×10^{-4}	3636590	1273699
6.037700	1.468070	515.4060	0.8867925	0.3285020×10^{-4}	3636590	1273699
6.037700	1.468070	289.3050	0.8759853	0.3285020×10^{-4}	3636590	1273699
6.037700	1.468070	1369.325	0.9945376	0.3285020×10^{-4}	3636590	1273699
6.037700	1.468070	320.7641	0.8818670	0.3285020×10^{-4}	3636590	1273699

Table 1.5: The first five samples of the $Z = Z_\odot$ model with the Bray kick. The weight field is the number of systems with these parameters we expect to observe per $10^6 M_\odot$ of star formation. $t_{\rm rej}$ and $t_{\rm evo}$ are the rejuvenation age and evolution age of the stellar model.

Chapter 2

Theoretical Background

Lives of great men all remind us; we can make our lives sublime; and, departing, leave behind us; footprints on the sands of time.

Henry Wadsworth Longfellow

We begin by working through the theory that underpins this work, starting from a very high-level understanding of stellar evolution, and moving on to deriving the equations of motion of a decaying binary system.

2.1 | Single Star Evolution: from formation to supernova

Stars spend the vast majority of their lifetimes on the "main sequence" – where they fuse hydrogen into helium in their cores. Stellar evolution can therefore be sorted into three stages, which we consider below.

2.1.1 | Pre-Main Sequence Evolution

Stars are born in rich H(II) regions in the universe, such as 30 Doradus, as clouds of gaseous hydrogen and helium. This amalgam of gas slowly grows in mass: when the cloud reaches the Jeans Mass $M_{\text{J}} = \frac{3k_{\text{B}}T}{GM}r$ (with r the radius of the gas cloud, M its mass, and T its temperature), the thermal energy of the cloud exceeds the gravitational energy binding it. In this regime, the gas cloud is unable to hold itself together in hydrostatic equilibrium, and thus collapses into a collection of many *protostars*. When an individual protostar's central temperature reaches 10^7 K, and hydrogen burning becomes dominant, the protostar has evolved onto the Main Sequence, and becomes a star in its own right.

2.1.2 | Main Sequence Evolution

Stars spend approximately 90% of their lifetime on the Main Sequence. When the star reaches Main Sequence evolution, it undergoes a period of hydrogen burning, where the core of the star converts hydrogen into helium, releasing energy via nuclear fusion. This energy

sustains the star until its fuel of hydrogen is exhausted, at which it begins a process of post-Main Sequence evolution.

2.1.3 | Post-Main Sequence Evolution

The evolution of a star primarily depends on its mass. For massive stars ($8M_\odot \lesssim M \lesssim 80M_\odot$), the star becomes a Red Giant, and undergoes helium burning, then carbon burning – converting helium to carbon to oxygen and neon via stellar nucleosynthesis. Strictly speaking, we should consider these stars to be formed of a hydrogen burning core and a helium burning shell (shell burning), though the precise modelling of a star that is not yet a compact remnant is beyond the scope of this thesis. This process continues, up to iron, nickel, and cobalt. During this period of burning, the star can enter one of three evolutionary channels (Meynet et al., 2011):

I: If the stellar mass is less than $30M_\odot$, it will become a Red Supergiant, Red Supergiants are the biggest stars in the Universe by volume, and have a spectral class of K or M (and thus appear red). An example of this type of star is Betelgeuse.

II: If the stellar mass is greater than $40M_\odot$, it will become a Luminous Blue Variable star (indicated by a strongly varying light curve, and are blue), and then a Wolf-Rayet star (indicated by the presence of helium in the spectra). This exact mechanism is contentious (see, for example, Groh et al., 2013, 2014), and no firm consensus has been reached as yet.

III: If the stellar mass is between 30 and 40 $M_\odot$, it *can* undergo all three stages: becoming a Red Supergiant, a Luminous Blue Variable, and eventually a Wolf-Rayet star. Which of the three categories it will enter into is dependent on $\dot{M}$, the rate of mass loss of the star.

We do not consider evolutionary channels which lead to White Dwarf formation, as LIGO does not release White Dwarf event rates as part of their transient catalogues.

After a period of time on one of the giant branches, the star dies in a supernova. This explosion renders the core of the star into a *compact remnant*, – a Black Hole or a Neutron Star, depending on the stellar mass. The process of forming this compact remnant is called *core collapse*, and the events themselves are called *core collapse supernovae* (CCSNe). Heger et al. (2003) estimates that this limit is $25M_\odot$: stars with final masses less than this form Neutron Stars, and stars with final masses greater than this form Black Holes. This limit is corroborated by Eldridge and Tout (2004).

Compact objects are extremely dense: a typical Neutron Star for example has a mass of about $1.4M_\odot$ and a radius of about 12 km (Lattimer, 2015). The density of the Neutron Star is therefore $\sim 3.85 \times 10^{17}$ kg m^{-3}, contrasted with $\sim$1406 kg m^{-3} for the Sun.

The process of stellar evolution is summarised in a flow chart in Fig. 2.3 on page 16.

The evolutionary process as described so far is complicated slightly, however, when the star is in a binary system.

2.2 | Binary Evolution

A binary star system is a pair of stars which orbit a common centre of mass. At any given time, we have an orbital configuration such as the one in Fig. 2.1, orbiting in the x-y plane. In practice this will have a component in the z direction, and an inclination relative to the observer (who sits along the z-axis). We will choose to reorient our coordinates such that the origin is at the barycentre of the system, and there is no z component. Binary orbits obey Kepler's Three Laws (see Appendix A for derivations).

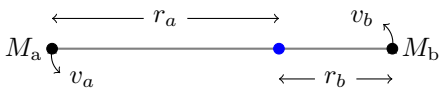


Figure 2.1: A schematic of a simple binary star system. The centre of mass is shown as a blue dot. Notice that since $r_a > r_b$, we have $v_a < v_b$ by Kepler's First Law, and $M_\mathrm{b} > M_\mathrm{a}$ by Newton's Second Law.

Most massive stars are not single, but exist in systems with multiplicity ≥ 2. The exact percentage is disputed among astrophysicists. Some authors, such as Duquennoy and Mayor (1991); Moe and Di Stefano (2017), claim that the percentage of stars which are in systems with two or more stars is $> 65\%$, however some authors such as Lada (2006) assert it to be as low as 33%. Moe and Di Stefano (2017) shows that multiplicity generally increases with mass, as in Table 1.3. Stellar evolution with n stars in the system is thus an n-body problem, as they orbit their centre of mass and undergo further evolution.

As the companion stars evolve along the main sequence, they may fill and eventually overflow their Roche Lobes, which are equipotential surfaces defined by the effective radius R_L, given in Eggleton (1983) by

$$\frac{R_\mathrm{L}}{a} \simeq \frac{0.49q^{2/3}}{0.6q^{2/3} + \ln\left(1 + q^{1/3}\right)} \tag{2.1}$$

where q is the mass ratio of the binary system and a is the semi-major axis. Eqn(2.1) defines a sphere with the same volume of the Roche Lobe. This approximation is accurate to $< 1\%$ for all $0 < q < \infty$, and models the Roche Lobe as a sphere. In reality, a Roche Lobe is actually tear-drop shaped (see Fig. 2.2), with the apex of the tear-drop being the inner Lagrangian point $\mathcal{L}_1$. A star being within its Roche Lobe means it does not transfer mass to its companion. However if a star overflows its Roche Lobe, the process of accretion must begin. We call this process *Roche Lobe overflow* (RLOF) and say the star has overfilled its Roche Lobe. Depending on the stability of the mass transfer, the two stars may enter an

epoch of Common Envelope Evolution (CEE) in which they evolve symbiotically, with the primary star contained within the low-density envelope of the secondary.

Stability of mass transfer depends on the magnitude of $\dot{M}$. If it is increasing rapidly, the mass transfer is said to be *unstable*. This typically occurs when the radius of the Roche Lobe shrinks faster than that of the star, indicating that the star is very rapidly losing mass.

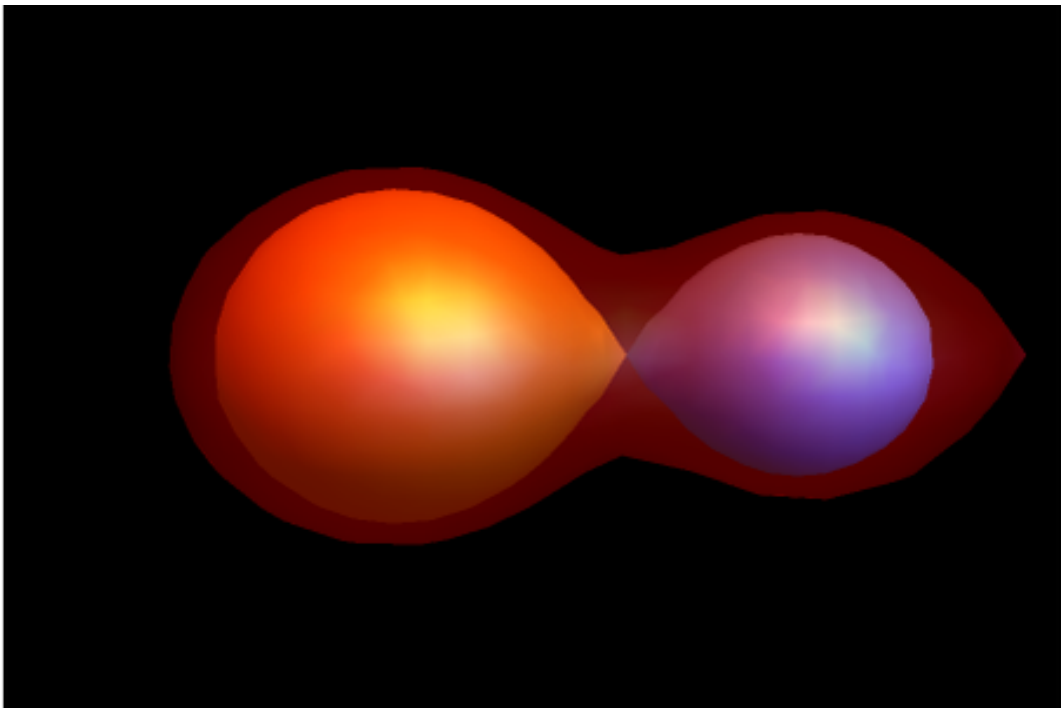


Figure 2.2: The Roche Lobe of a sample binary star system. In this image, the orange and purple surfaces are the Roche Lobes of the primary and secondary stars respectively. The thinner red cloud around both is the common envelope. Image by Jeff Bryant, from https://demonstrations.wolfram.com/ContactBinaryStarEnvelopes/, under CC BY-NC-SA license.

Once the two stars have become compact remnants, they begin the inspiral phase of their evolution. This occurs as they lose angular momentum due to the emission of gravitational waves – and thus their orbits shrink. At the end of this phase is the merger, which is the signal detectable by LIGO.

2.2.1 | Natal supernova kicks

The supernova often imparts some additional velocity onto the compact remnant: we call this a *supernova kick*. Although these kicks were known to exist as early as 1970 (Gunn and Ostriker, 1970), the precise nature of this kick is not yet fully understood. There are two kicks we consider here. The "Hobbs Kick" (Hobbs et al., 2005) follows a Maxwellian

distribution (denoted by $P(v_k)$) with $\sigma = 265$ km/s, namely (from Martin et al. 2009):

$$P(v_\mathrm{k})\,\mathrm{d}v_\mathrm{k} = \sqrt{\frac{2}{\pi}}\frac{v_\mathrm{k}^2}{\sigma^3}\exp\left(-\frac{v_\mathrm{k}^2}{2\sigma^2}\right)\mathrm{d}v_\mathrm{k} \tag{2.2}$$

whereas the "Bray Kick" (Bray, 2018) is linear in the ejecta-remnant mass ratio:

$$v_\mathrm{kick} = \alpha\left(\frac{M_\mathrm{ejecta}}{M_\mathrm{remnant}}\right) + \beta \tag{2.3}$$

Bray (2018) finds that the best-fitting parameters α, β are those given in Table 2.1. It is pertinent to note the large uncertainties in the parameter β; part of the work of this thesis is to refine these parameters to more acceptable levels.

For the Hobbs kick, both velocity and kick direction are sampled randomly (with velocity sampled from eqn(2.2) and direction sampled uniformly over all possible directions). On the other hand, the Bray kick takes the velocity as given by 2.3, and samples the direction randomly. This effective random sampling instead of preferentially aligning kick direction with the spin axis is justified by Andrews and Mandel (2019) and reinforced by Tauris et al. (2017).

This random sampling produces a range of possible evolution pathways for the system after the first supernova. In extreme cases, it may cause the binary to become unbound.

Bray (2018) concludes with the remark that if their work is correct, then we may expect significantly more DNS detections in the then-impending LIGO O3 data release.

Model	α	β
2D	100^{+30}_{-20}	-170^{+100}_{-100}
3D	60^{+30}_{-10}	130^{+10}_{-200}

Table 2.1: The best-fitting paramters α, β for the Bray kick. The values chosen here replicate the earlier work of Hobbs et al. (2005), and in later chapters we shall refine the uncertainties in the fit parameters, especially that of β.

2.2.2 | Decaying binaries

We start with an input file containing the initial conditions of the binary – masses, semimajor axes, and eccentricities. From this, we must determine the binary's evolution and decay due to gravitational radiation. This emission will lead to the circularisation of the binary's orbit, a decrease of the period of the binary, and a merger. To that end, we seek an expression for $\frac{\mathrm{d}a}{\mathrm{d}t}$ and $\frac{\mathrm{d}e}{\mathrm{d}t}$ – the two parameters which describe the evolution of a binary system throughout time. The derivation below broadly follows the original work of Peters (1964), expanded on by Nyadzani and Razzaque (2020), though small alterations have been made to ensure that

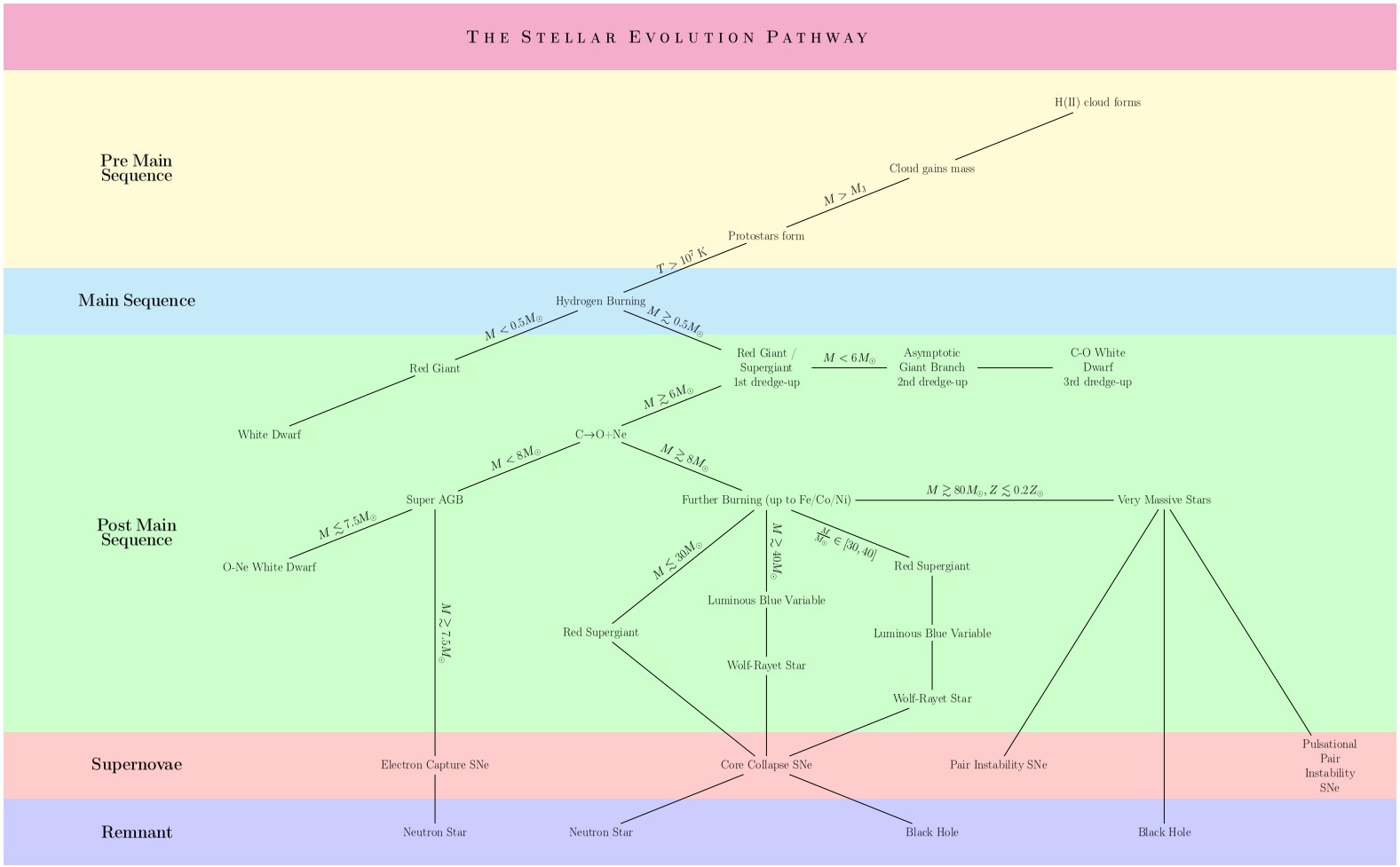


Figure 2.3: A flow chart representing the process of single star evolution. Notice that almost all stars between 7.5 and 80 $M_\odot$ produce a compact remnant. Binaries complicate this evolution. Image is made by author, but based on Eldridge (2020).

the metric signature and notation is aligned with this thesis. BPASS only produces the initial parameters for the compact remnant binary. From that point, they decay via gravitational radiation, and may be approximated as two point masses. Thus, to compute the merger rate densities and the period-eccentricity distributions, we must derive expressions for the future evolution of the binary system.

Working in the weak-field limit, we write the metric as

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu} \tag{2.4}$$

with η the Minkowski metric, $|h| \ll 1$, and $\kappa^2 = 32\pi G$. This metric induces (see, among others, Einstein, 1916; Peters, 1964) the wave equation

$$\Box \overline{h}_{\mu\nu} = -\frac{1}{2}\kappa T_{\mu\nu} \tag{2.5}$$

Given some plane wave solution to this wave equation $h_{\mu\nu} = Ae_{\mu\nu}\cos(\mathbf{k}\cdot\mathbf{x} - \omega t)$ (with $e_{\mu\nu}$ a unit polarization tensor), we have the energy density

$$U = \frac{1}{2}c^2\omega^2 A^2 \tag{2.6}$$

The two possible non-zero spacelike and transverse polarizations for a wave travelling in the z-direction are

$$e_1 = \frac{1}{\sqrt{2}}(\hat{x}\hat{x} - \hat{y}\hat{y}) \qquad e_2 = \frac{1}{\sqrt{2}}(\hat{x}\hat{y} - \hat{y}\hat{x}) \tag{2.7}$$

where we denote the "plus" polarisation as e_1 and the "cross" polarisation as e_2. One soon finds that the power P radiated into a solid angle Ω is given by

$$\frac{\mathrm{d}P}{\mathrm{d}\Omega} = \frac{G}{8\pi c^5}\left(\frac{\partial^3 Q_{ij}}{\partial t^3}e_{ij}\right)^2 \tag{2.8}$$

with

$$Q_{ij} = m^\alpha x_{\alpha i} x_{\alpha j} \tag{2.9}$$

The total power is obtained reasonably straightforwardly:

$$\begin{aligned}\frac{\mathrm{d}P}{\mathrm{d}\Omega} = \frac{G}{8\pi c^5}\Bigg[&\left(\frac{\mathrm{d}^3 Q_{ij}}{\mathrm{d}t^3}\right)^2 - 2n_i\frac{\mathrm{d}^3 Q_{ij}}{\mathrm{d}t^3}n_k\frac{\mathrm{d}^3 Q_{kj}}{\mathrm{d}t^3} - \frac{1}{2}\left(\frac{\mathrm{d}^3 Q_{ii}}{\mathrm{d}t^3}\right)^2 \\ &+\frac{1}{2}\left(n_i n_j\frac{\mathrm{d}^3 Q_{ij}}{\mathrm{d}t^3}\right)^2 + \frac{\mathrm{d}^3 Q_{ii}}{\mathrm{d}t^3}n_j n_k\frac{\mathrm{d}^3 Q_{jk}}{\mathrm{d}t^3}\Bigg]\end{aligned} \tag{2.10}$$

with $\hat{n}$ a unit vector in the direction of radiation. The total power is then found by

integrating this equation over all directions:

$$P = \frac{G}{5c^5}\left(\frac{\mathrm{d}^3 Q_{ij}}{\mathrm{d}t^3}\frac{\mathrm{d}^3 Q_{ij}}{\mathrm{d}t^3} - \frac{1}{3}\frac{\mathrm{d}^3 Q_{ii}}{\mathrm{d}t^3}\frac{\mathrm{d}^3 Q_{jj}}{\mathrm{d}t^3}\right) \tag{2.11}$$

This result has all been very general, and we are now in a position to inject the physics. Suppose the primary star has mass M and the secondary mass m. In the xy-plane, with the origin at the center of mass, we can parameterise the positions of the two stars as:

$$\mathbf{x}_1 = \begin{pmatrix} d_1 \cos\varphi \\ d_1 \sin\varphi \end{pmatrix} \qquad \mathbf{x}_2 = \begin{pmatrix} -d_2 \cos\varphi \\ -d_2 \sin\varphi \end{pmatrix} \tag{2.12}$$

with

$$d_1 = \frac{m}{m+M}d \qquad d_2 = \frac{M}{m+M}d \tag{2.13}$$

The components of the tensor Q_{ij} are

$$Q_{xx} = \mu d^2 \cos^2\varphi \tag{2.14}$$

$$Q_{yy} = \mu d^2 \sin^2\varphi \tag{2.15}$$

$$Q_{xy} = Q_{yx} = \mu d^2 \sin\varphi\cos\varphi \tag{2.16}$$

with the rest vanishing, and with μ as the reduced mass. Now, as binary stars obey Kepler's laws, Kepler's orbit equation holds (see Section A.1 for the proof):

$$d = \frac{a(1-e^2)}{1+e\cos\phi} \tag{2.17}$$

whilst the angular velocity is given by

$$\phi = \frac{\sqrt{G(m+M)a(1-e^2)}}{d^2} \tag{2.18}$$

We therefore have

$$\frac{\mathrm{d}^3 Q_{xx}}{\mathrm{d}t^3} = \gamma(1+e\cos\phi)^2(2\sin(2\phi) + 3e\sin\phi\cos^2\phi) \tag{2.19}$$

$$\frac{\mathrm{d}^3 Q_{yy}}{\mathrm{d}t^3} = -\gamma(1+e\cos\phi)^2\big(2\sin(2\phi) + e\sin\phi(1+3\cos^2\phi)\big) \tag{2.20}$$

$$\frac{\mathrm{d}^3 Q_{xy}}{\mathrm{d}t^3} = \frac{\mathrm{d}^3 Q_{yx}}{\mathrm{d}t^3} = -\gamma(1+e\cos\phi)^2\big(2\cos(2\phi) - e\cos\phi(1-3\cos^2\phi)\big) \tag{2.21}$$

where the quantity γ is defined as

$$\gamma^2 = \frac{4G^3m^2M^2(m+M)}{a^5(1-e^2)^5} \tag{2.22}$$

The total power is therefore

$$P = \frac{8}{15}\frac{G^4}{c^5}\frac{m^2M^2(m+M)}{a(1-e^2)^5}(1+e\cos\varphi)^4\big(12(1+e\cos\varphi)^2+e^2\sin^2\varphi\big) \tag{2.23}$$

Averaging this over one orbit yields the average power, which we write now as $\left\langle \frac{\mathrm{d}E}{\mathrm{d}t} \right\rangle$ in line with Nyadzani and Razzaque (2020),

$$\left\langle \frac{\mathrm{d}E}{\mathrm{d}t} \right\rangle = -\frac{32}{5}\frac{G^4}{c^5}\frac{m^2M^2(m+M)}{a^5(1-e^2)^{\frac{7}{2}}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \tag{2.24}$$

The average angular momentum is

$$\left\langle \frac{\mathrm{d}L}{\mathrm{d}t} \right\rangle = -\frac{32}{5}\frac{G^{\frac{7}{2}}m^2M^2\sqrt{m+M}}{c^5a^{\frac{7}{2}}(1-e^2)^2}\left(1+\frac{7}{8}e^2\right) \tag{2.25}$$

We now write the gravitational energy from the two-body problem:

$$E = -\frac{GMm}{2a} \tag{2.26}$$

$$\implies \frac{\mathrm{d}E}{\mathrm{d}t} = \frac{GMm}{2}\frac{1}{a^2}\frac{\mathrm{d}a}{\mathrm{d}t} \tag{2.27}$$

Define the constant β as

$$\beta \equiv \frac{64}{5}\frac{G^3mM(m+M)}{c^5} \tag{2.28}$$

Now insert eqn(2.27) into eqn(2.24) to obtain

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{2a^2}{GMm}\frac{32}{5}\frac{G^4m^2M^2(m+M)}{c^5a^5(1-e^2)^{\frac{7}{2}}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \tag{2.29}$$

$$= -\frac{\beta}{a^3(1-e^2)^{\frac{7}{2}}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \tag{2.30}$$

The derivation for $\frac{\mathrm{d}e}{\mathrm{d}t}$ is a little more involved. The angular momentum from the two-body problem (see eqn(A.15) is

$$L^2 = \frac{GM^2m^2a\big(1-e^2\big)}{m+M} \tag{2.31}$$

$$\implies 2L\frac{\mathrm{d}L}{\mathrm{d}t} = \frac{Gm^2M^2}{m+M}\frac{\mathrm{d}}{\mathrm{d}t}\big(a(1-e^2)\big) \tag{2.32}$$

$$= \frac{Gm^2M^2}{m+M}\left[\frac{\mathrm{d}a}{\mathrm{d}t}\big(1-e^2\big) - 2ae\frac{\mathrm{d}e}{\mathrm{d}t}\right] \tag{2.33}$$

Multiplying each side of eqn(2.25) by $2L$ and substituting this in, we obtain

$$\frac{Gm^2M^2}{m+M}\left[\frac{\mathrm{d}a}{\mathrm{d}t}\big(1-e^2\big) - 2ae\frac{\mathrm{d}e}{\mathrm{d}t}\right] = -L\frac{64}{5}\frac{G^{\frac{7}{2}}m^2M^2\sqrt{m+M}}{c^5a^{\frac{7}{2}}\big(1-e^2\big)^2}\left(1+\frac{7}{8}e^2\right) \tag{2.34}$$

$$\frac{1}{m+M}\left[\frac{\mathrm{d}a}{\mathrm{d}t}\big(1-e^2\big) - 2ae\frac{\mathrm{d}e}{\mathrm{d}t}\right] = -\sqrt{\frac{GM^2m^2a(1-e^2)}{m+M}}\frac{64}{5}\frac{G^{\frac{5}{2}}\sqrt{m+M}}{c^5a^{\frac{7}{2}}\big(1-e^2\big)^2}\left(1+\frac{7}{8}e^2\right) \tag{2.35}$$

$$\frac{\mathrm{d}a}{\mathrm{d}t}\big(1-e^2\big) - 2ae\frac{\mathrm{d}e}{\mathrm{d}t} = -\frac{\beta}{a^3\big(1-e^2\big)^{\frac{3}{2}}}\left(1+\frac{7}{8}e^2\right) \tag{2.36}$$

$$2ae\frac{\mathrm{d}e}{\mathrm{d}t} = \frac{\beta}{a^3\big(1-e^2\big)^{\frac{3}{2}}}\left(1+\frac{7}{8}e^2\right) + \frac{\mathrm{d}a}{\mathrm{d}t}\big(1-e^2\big) \tag{2.37}$$

$$2ae\frac{\mathrm{d}e}{\mathrm{d}t} = \frac{\beta}{a^3\big(1-e^2\big)^{\frac{3}{2}}}\left(1+\frac{7}{8}e^2\right) - \frac{\beta}{a^3\big(1-e^2\big)^{\frac{5}{2}}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \tag{2.38}$$

$$\frac{\mathrm{d}e}{\mathrm{d}t} = \frac{\beta}{a^4\big(1-e^2\big)^{\frac{5}{2}}}\frac{\big(1-e^2\big)\big(1+\frac{7}{8}e^2\big) - \big(1+\frac{73}{24}e^2+\frac{37}{96}e^4\big)}{2e} \tag{2.39}$$

$$= \frac{\beta}{a^4\big(1-e^2\big)^{\frac{5}{2}}}\left[-\frac{19}{12}e^3 - \frac{121}{192}e\right] \tag{2.40}$$

$$= -\frac{19}{12}\frac{\beta}{a^4\big(1-e^2\big)^{\frac{5}{2}}}\left(e+\frac{121}{304}e^3\right) \tag{2.41}$$

Hence we arrive at the coupled differential equations describing the decay of a binary star system:

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \frac{-\beta}{a^3(1-e^2)^{\frac{7}{2}}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \tag{2.42a}$$

$$\frac{\mathrm{d}e}{\mathrm{d}t} = -\frac{19}{12}\frac{\beta}{a^4(1-e^2)^{\frac{5}{2}}}\left(e+\frac{121}{304}e^3\right) \tag{2.42b}$$

The coalescence time is by definition the time it takes for the binary system to merge after the secondary supernova. We can step towards this in a similar way to Peters (1964). First, dividing eqn(2.42a) by eqn(2.42b) gives

$$\frac{\mathrm{d}a}{\mathrm{d}e} = \frac{12}{19}\frac{a}{e}\frac{1+\frac{73}{24}e^2+\frac{37}{96}e^4}{(1-e^2)\left(1+\frac{121}{304}e^2\right)} \tag{2.43}$$

We can integrate this however we choose to obtain $a(e)$, namely

$$a(e) = \frac{c_0 e^{\frac{12}{19}}}{(1-e^2)}\left(1+\frac{121}{304}e^2\right)^{\frac{870}{2299}} \tag{2.44}$$

with

$$c_0 = a_0\left(1-e_0^2\right)e_0^{-\frac{12}{19}}\left(1+\frac{121}{304}e_0^2\right)^{-\frac{870}{2299}} \tag{2.45}$$

Now, substituting eqn(2.44) into eqn(2.42b), we see

$$\frac{\mathrm{d}e}{\mathrm{d}t} = -\frac{19}{12}\frac{\beta}{a^4(1-e^2)^{\frac{5}{2}}}\left(e+\frac{121}{304}e^3\right) \tag{2.46}$$

$$= -\frac{19}{12}\frac{\beta}{\left(\frac{c_0 e^{\frac{12}{19}}}{(1-e^2)}\left(1+\frac{121}{304}e^2\right)^{\frac{870}{2299}}\right)^4(1-e^2)^{\frac{5}{2}}}\left(e+\frac{121}{304}e^3\right) \tag{2.47}$$

$$= -\frac{19}{12}\frac{\beta}{c_0^4}\frac{\left(1-e^2\right)^{\frac{3}{2}}e}{e^{\frac{48}{19}}}\frac{\left(1+\frac{121}{304}e^2\right)}{\left(1+\frac{121}{304}e^2\right)^{\frac{3480}{2299}}} \tag{2.48}$$

$$= -\frac{19}{12}\frac{\beta}{c_0^4}\frac{\left(1-e^2\right)^{\frac{3}{2}}e^{-\frac{29}{19}}}{\left(1+\frac{121}{304}e^2\right)^{\frac{1181}{2299}}} \tag{2.49}$$

$$\tag{2.50}$$

Now write this as $\frac{\mathrm{d}t}{\mathrm{d}e}$ and integrate to get the coalescence time:

$$\frac{\mathrm{d}t}{\mathrm{d}e} = -\frac{12}{19}\frac{c_0^4}{\beta}\frac{\left(1+\frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}e^{-\frac{29}{19}}} \tag{2.51}$$

$$T_\mathrm{c} = \int_0^{T_\mathrm{c}} \mathrm{d}t = -\frac{12}{19}\frac{c_0^4}{\beta}\int_{e_0}^{0}\frac{\left(1+\frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}e^{-\frac{29}{19}}}\,\mathrm{d}e \tag{2.52}$$

$$= \frac{12}{19}\frac{c_0^4}{\beta}\int_{0}^{e_0}\frac{\left(1+\frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}e^{-\frac{29}{19}}}\,\mathrm{d}e \tag{2.53}$$

i.e.,

$$T_\mathrm{c}(a_0, e_0) = \frac{12}{19}\frac{c_0^4}{\beta}\int_{0}^{e_0}\frac{e^{\frac{22}{19}}\left(1+\frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}}\,\mathrm{d}e \tag{2.54}$$

If $e_0 = 0$ then the orbit is circular and the coalescence time is given by

$$T_\mathrm{c} = \frac{a_0^4}{4\beta} \tag{2.55}$$

If the orbit is not circular, the canonical[1] way to solve this is to set up three cases:

Case I: *Small eccentricities* ($e_0 < 0.01$).

In this regime, we can use another equation from Peters (1964) to approximate the coalescence time;

$$T_\mathrm{c} \simeq \frac{c_0^4}{4\beta}e_0^{\frac{48}{19}} \tag{2.56}$$

Case II: *Intermediate eccentricities* ($0.01 < e_0 < 0.99$).

In this regime we have no choice but to integrate eqn(2.54) numerically. We use a simple Euler integrator, as is implemented in Barrett et al. (2016) with a step size of $\mathrm{d}t = \frac{e_0}{10000}$, to solve this integral.

Case III: *High eccentricities* ($e_0 > 0.99$).

We can use another approximation from Peters (1964), namely;

$$T_\mathrm{c} \simeq \frac{768}{425}\tau_\mathrm{circ}\left(1-e_0^2\right)^{\frac{7}{2}} \tag{2.57}$$

where τ_circ is the coalescence time for an orbit with the same radius as the semimajor axis of the binary. This formalism presents clean ways to approximate the coalescence time.

[1] See, among others, Peters (1964); Barrett et al. (2016); Nyadzani and Razzaque (2020)

The results of this numerical integration are shown in Fig. 2.4, and we can see that the result is smooth for all e_0.

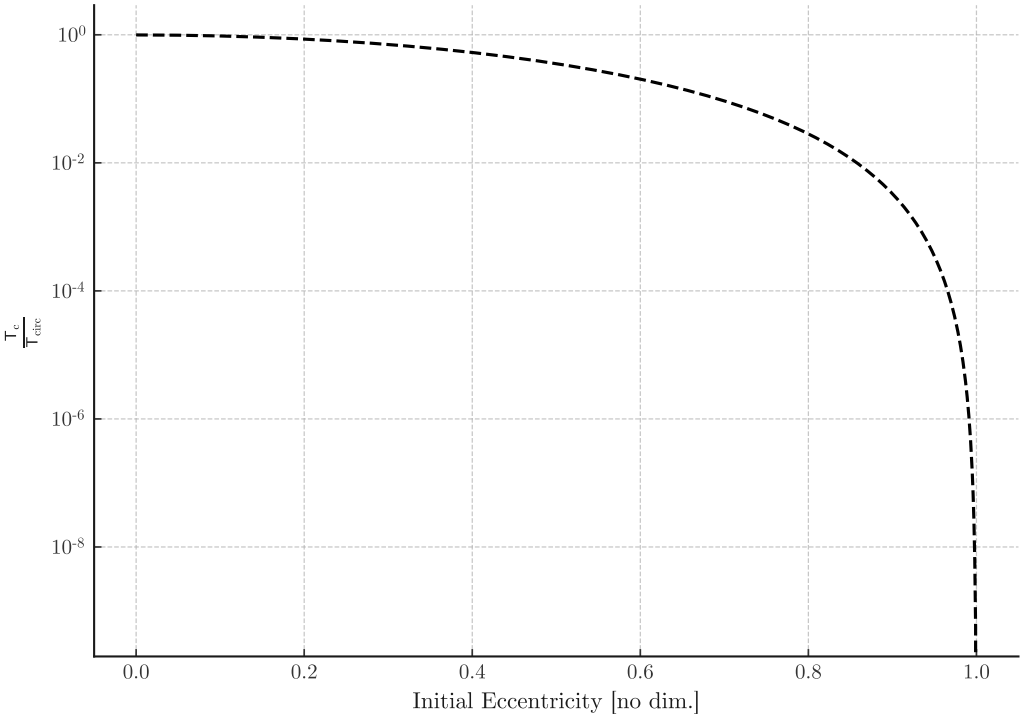


Figure 2.4: The coalescence time of a binary with parameters $m_1 = 1.4M_\odot, m_2 = 1.4M_\odot, P_0 = 100$ days, and initial eccentricity e_0. We take the evolution + rejuvenation time to be 2×10^6 years. Note that we compute $\frac{T_c}{T_{\rm circ}}$, i.e. the ratio of the coalescence time to that of a circular binary with the same parameters. This is to remove some dependence on the initial separation and evolution + rejuvenation time, which can be essentially arbitrarily chosen.

In particular, for numerical integration we use an Euler integrator, where

$$a_n = a_{n-1} + h_n \frac{\mathrm{d}a}{\mathrm{d}t}$$
$$e_n = e_{n-1} + h_n \frac{\mathrm{d}e}{\mathrm{d}t}$$

with adaptive step size h_n. If we impose sufficiently strong cutoff conditions – we select a 10km semimajor axis or a maximum total integration time of 10^{11} years – then the coalescence time can be written as

$$T_{\rm c} = \sum_n h_n \tag{2.58}$$

i.e. we can simply sum the step sizes to resolve the coalescence time.

2.2.3 | Merge Rates & Delay-Time Distributions

Now that we know how binary stars evolve, given some time t, we can compute the merge rate $m(t)$ of an ensemble of binary star systems by using the following heuristic:

$$m(t) = \#\text{ of binaries for which } t > T_{\rm c} + t_{\rm rej} + t_{\rm evo}$$

where $t_{\rm rej}$ is the rejuvenation age and $t_{\rm evo}$ is the evolution age. Of course, the merge rate then depends on t, leading to the potential of a rate distribution depending wholly on the value of t we select. We compute this distribution by considering the merge rate following an instantaneous burst of star formation at time $t = 0$; this is called the Delay-Time Distribution (DTD) of the system.

An example of such a distribution for a small ensemble of stars is shown in Fig. 2.5. There are several inferences we can make from this result. Firstly, the nature of the kick has an appreciable impact on the DTD – the peak of the DTD changes subtly for each model. The Bray kick produces fewer mergers at earlier lookback times, but more events at later times. As the form of the kick changes the DTD, it will also change the event rate distribution. We can conclude from this DTD that the Bray kick leads to wider and/or lower eccentricity binaries, which take longer to merge, whereas the Hobbs kick has more eccentric or short period binaries, which merge more rapidly.

2.2.4 | Stellar Formation History

The analysis we have performed so far assumes that all stars are created at the beginning of the Universe, and no stars are born over time. This is clearly an improper assumption to make, and we cannot take the delay-time distribution as an event rate plot as a consequence. To account for stellar evolution, we must instead consider the *stellar formation history* (SFH) of the Universe. For our purposes this is introduced in Madau and Dickinson (2014) via the empirically-derived expression for the Stellar Formation Rate Density (SFRD), $\psi(z)$, where z is redshift, namely

$$\psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}}\ \mathrm{M}_\odot\ \mathrm{yr}^{-1}\ \mathrm{Mpc}^{-3} \tag{2.59}$$

A plot of SFRD against Lookback Time (the amount of time since the event occurred until now) is given in Fig. 2.6.

We now wish to compute the star formation *history*. To do so, we must convert the *lookback time* $t_{\rm L}$ under consideration into a redshift. This computation is not possible analytically and must be done via a root-finding method. As outlined in Hogg (2000), the

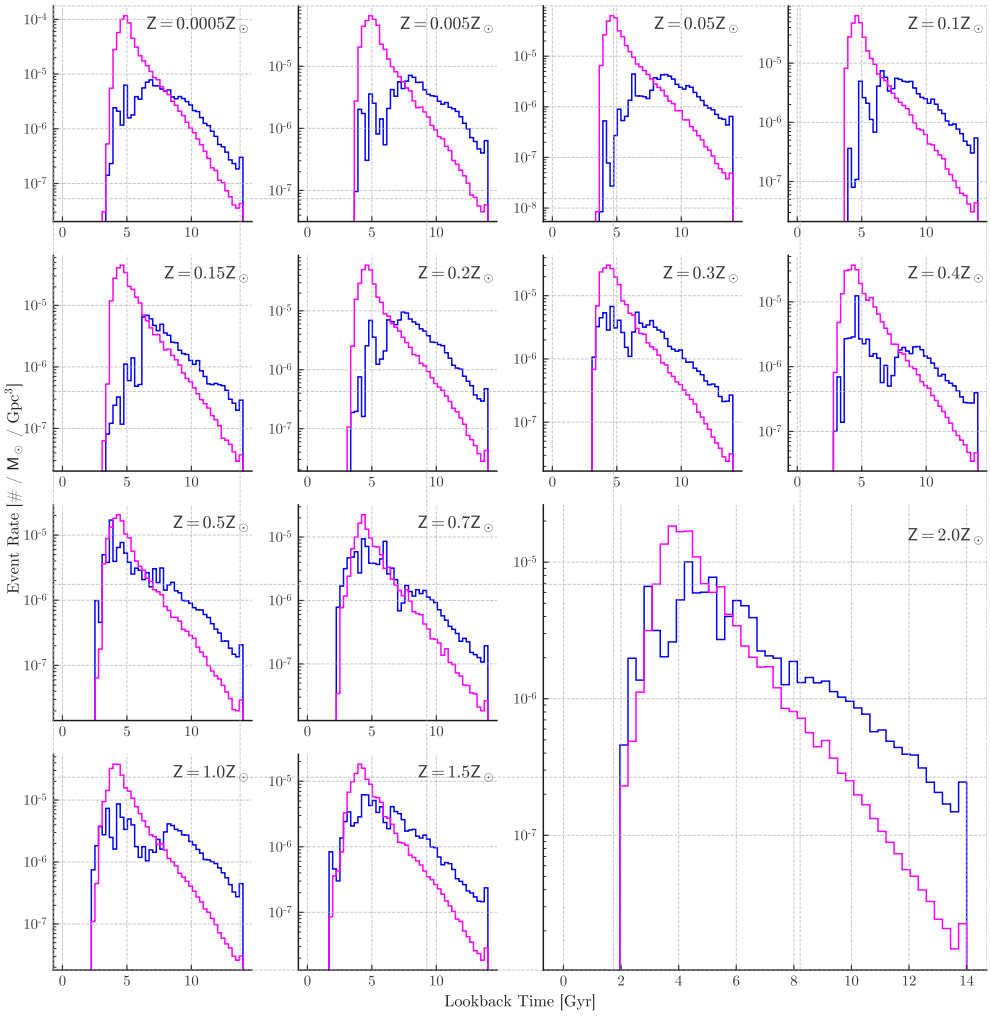


Figure 2.5: Delay time distribution for a small sample of stars, with varying metallicity. The magenta histogram is data from the Hobbs kick, whereas the blue histogram is data from the Bray kick. The data was generated by Tui, a script developed by Assoc. Prof. Eldridge at the University of Auckland, and a core part of the BPASS suite.

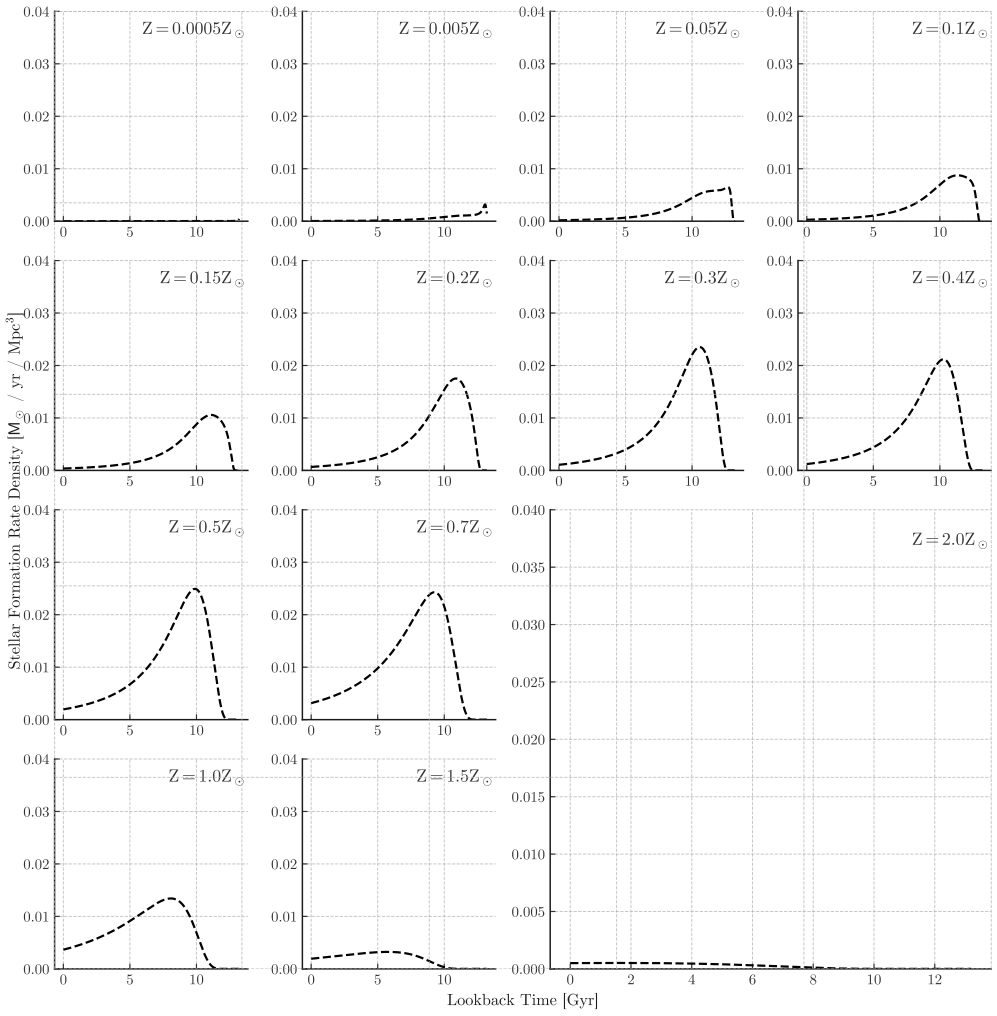


Figure 2.6: The Stellar Formation Rate Density function, broken down by BPASS metallicity, as a function of Lookback Time. The SFRD peaks at $t_{\rm L} \simeq 10$ Gyr, which is approximately $z = 2$.

equation for lookback time *in terms of* redshift is

$$t_{\mathrm{L}} = t_{\mathrm{H}} \int_0^z \frac{\mathrm{d}z'}{(1+z')E(z')} \tag{2.60}$$

where $t_{\mathrm{H}} \equiv \frac{1}{H_0}$ is the Hubble time, and $E(z')$ is defined by

$$E(z) \equiv \sqrt{\Omega_{\mathrm{M}}(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda} \tag{2.61}$$

where $\Omega_{\mathrm{M}}, \Omega_k$, and Ω_Λ are dimensionless parameters quantifying the matter density, curvature of space, and cosmological constant respectively. An added constraint is that the first Friedmann relation holds, namely that

$$\Omega_{\mathrm{M}} + \Omega_k + \Omega_\Lambda = 1 \tag{2.62}$$

To solve eqn(2.60) for the redshift, we must minimise the difference between the LHS and RHS: i.e. the redshift z_{L} corresponding to lookback time t_{L} is

$$z_{\mathrm{L}} = \min \left| t_{\mathrm{L}} - t_{\mathrm{H}} \int_0^z \frac{\mathrm{d}z'}{(1+z')E(z')} \right| \tag{2.63}$$

where the minimisation is performed over z. If we assign the transformation $(t_1, t_2) \mapsto (z_1, z_2)$, then the SFH χ in the interval (t_1, t_2) is given by

$$\chi \equiv \int_{z_1}^{z_2} \psi(z)\,\mathrm{d}z = 0.015 \int_{z_1}^{z_2} \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}}\,\mathrm{d}z \tag{2.64}$$

Analytic solutions to this equation do exist, but are defined in terms of hypergeometric functions and Pochhammer symbols. It is therefore computationally easier to perform the integral numerically.

2.2.5 | Mixed-Metallicity Models

Our analysis so far has assumed the Universe has a homogenous distribution of metallicity – that all stars have a constant metallicity. Of course this is not true, as some stars are metal-rich and some are metal-poor, generally depending on when they were formed. To incorporate the variation in metallicity, we introduce the *fractional mass density* (FMD) factor Ψ, as derived in Langer and Norman (2006), defined by

$$\Psi\left(z, \frac{Z}{Z_\odot}\right) = \frac{\hat{\Gamma}\left(\alpha+2, \left(\frac{Z}{Z_\odot}\right)^\beta 10^{0.15\beta z}\right)}{\Gamma(\alpha+2)} \tag{2.65}$$

which assumes that the mass-metallicity relation of the universe is

$$\frac{M}{M_*} = K\left(\frac{Z}{Z_\odot}\right)^\beta \tag{2.66}$$

We adopt the same fiducial values as Langer and Norman (2006) – $\alpha = -1.16, \beta = 2, K = 1$. Notably, $K = 1$ is what allows us to use a scaling of $\frac{\mathrm{d}[Z]}{\mathrm{d}t} = -0.15$ dex per unit redshift; a discussion of the implications of this is found in Langer and Norman (2006).

This gives us a modified SFRD of

$$\Psi\left(z, \frac{Z}{Z_\odot}\right) = \psi(z)\frac{\hat{\Gamma}\left[0.84, (Z/Z_\odot)^2 10^{0.3z}\right]}{\Gamma(0.84)} \tag{2.67}$$

where $\hat{\Gamma}$ and Γ are the (upper) incomplete and complete Gamma functions respectively. Ψ is a culmulative function, however, and the SFH χ will thus include contributions at both at metallicity Z, and also at all metallicities $Z' < Z$ (which are given in Table 1.4). Thus, computationally, we select $Z_\mathrm{c} = \frac{Z_{i-1}+Z_i}{2}$ (with $Z_0 = 0$) as our metallicity bins, and subtract the values of all previous bins to obtain the value of χ at metallicity bin Z and Z alone. We, effectively, are weighting the SFRD to "prefer" certain metallicities, and this is summarised in Table 1.4 and Fig. 2.6.

Formally, weighting the SFRD in this way means that

$$\begin{aligned}
\chi(z, Z) &= \int_{z_1}^{z_2} \Psi\left(z, \frac{Z}{Z_\odot}\right) \mathrm{d}z \\
&= \int_{z_1}^{z_2} \psi(z)\frac{\hat{\Gamma}\left[0.84, (Z/Z_\odot)^2 10^{0.3z}\right]}{\Gamma(0.84)} \mathrm{d}z \\
&= 0.015 \int_{z_1}^{z_2} \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \frac{\hat{\Gamma}\left[0.84, (Z/Z_\odot)^2 10^{0.3z}\right]}{\Gamma(0.84)} \mathrm{d}z
\end{aligned}$$

Again, this has no exact analytical solution – thus we numerically integrate it.

2.2.6 | Event Rates at $z = 0$ and beyond

Having defined the delay time distribution $m(t)$ and SFH $\chi(z, Z)$, we are now in a position to compute the event rate $\mathcal{R}$, which is the convolution of m with χ:

$$\mathcal{R}(z) = \sum_Z m(z) * \chi(z, Z) \tag{2.68}$$

And the event rate today, at $z = t_\mathrm{L} = 0$, is the content of the first redshift bin, i.e.,

$$\mathcal{R}_0 \equiv \mathcal{R}(0) \tag{2.69}$$

This in hand, we can compute the event rate densities using the selection of DTDs in Fig. 2.5 and SFRDs in Fig. 2.6. The result is given in Fig. 2.7. We see immediately that the Bray kick produces an event rate that is concordant with the upper uncertainty bound, but not the lower one or the nominal value. Note also that the rate that we report is lower than Eldridge et al. (2018), this is due to refinements within Tui around the rejuvenation age calculation.

Another important observation from Fig. 2.7 is that the peak event rate is different for each model. The peak event rate for the Bray kick is approximately 7 Gyr ago, whereas the peak for the Hobbs kick is approximately 9.5 Gyr ago. This has an appreciable impact on the distribution and influence of NSNS systems in the universe, as discussed in Rose et al. (2021).

2.3 | Methodology

The primary work of this thesis involves turning the prior details into a Python package named Takahe.[2] Takahe's code is available on GitHub[3] and the event rate code in particular is given in Appendix B. Takahe falls within the BPASS suite (see, for example, Eldridge et al. (2017); Stevance et al. (2020) for further discussion on the BPASS suite) of astronomical research. Takahe is partially written in the Julia programming language (Bezanson et al., 2014) to achieve speed gains.

Takahe is a library to propagate binaries in time according to the evolution equations eqn(2.42). The BPASS models are propagated through Tui, and then used as input to Takahe, which computes the merger rate density of the given ensemble of data, factoring in the SFR (see Chapter 3). Takahe is also capable of computing the period-eccentricity distributions of these datafiles (see Chapter 4). We then compare these results against observational data, such as Table 1.1 on page 4 for the merger rates, and Table 1.2 on page 6 for the period-eccentricity distributions.

As discussed previously, Takahe uses an Euler integrator to solve eqn(2.42). The adaptive timestep is selected via

$$h_n = \min\left\{\frac{10^{6.05} - 10^{5.95}}{2}t, \frac{a}{10\,\mathrm{d}a}, \frac{e}{10\,\mathrm{d}e}\right\} \tag{2.70}$$

This selects a timestep in the best direction: minimising either da or de. The first parameter serves to force the minimum timestep to be half the width of the smallest BPASS time bin – thus, each iteration of the integrator will at most step through one BPASS time bin.

Takahe's implementation of the event rate calculation is given in Appendix B. The plots

[2] The Takahe is an endangered New Zealand bird, which is viewed as taonga to the Ngāi Tahu iwi

[3] https://github.com/krytic/takahe

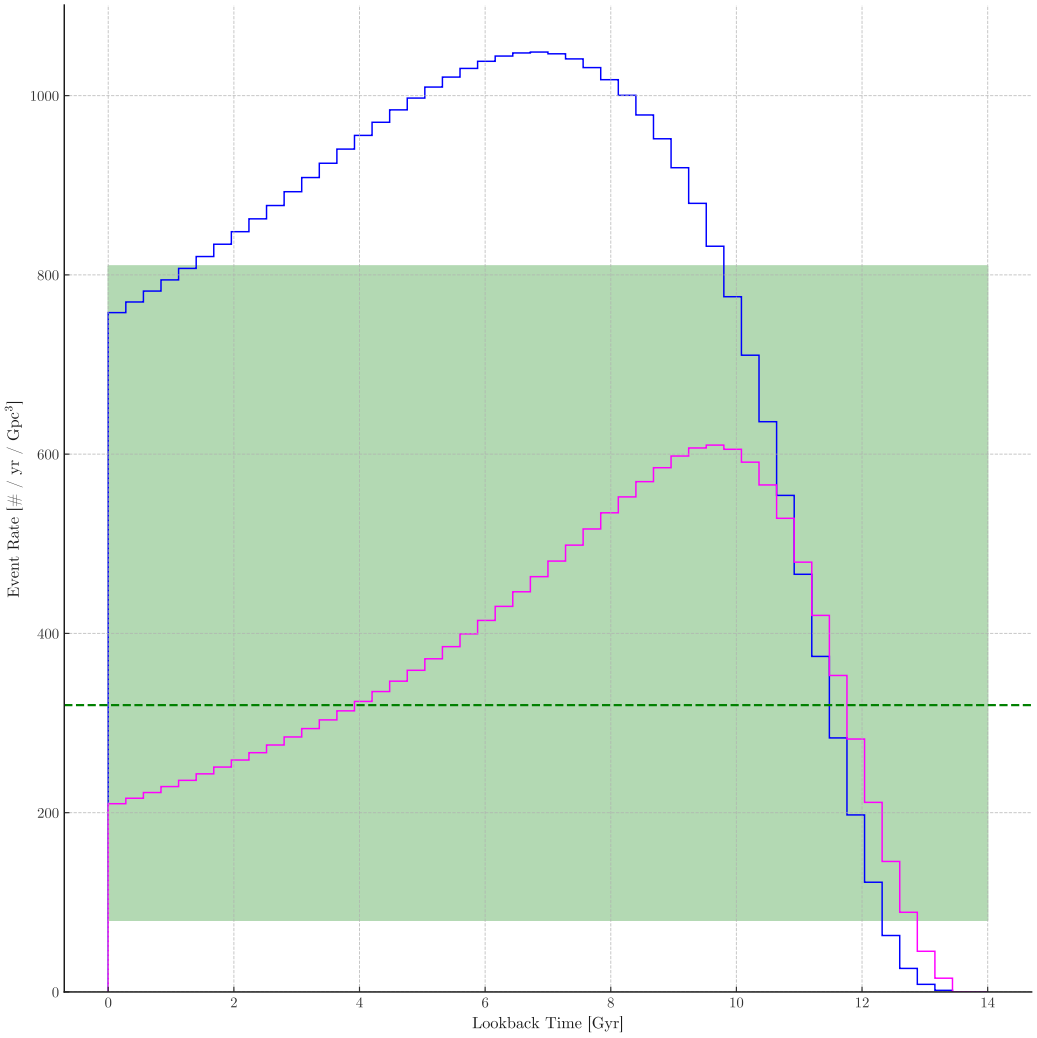


Figure 2.7: The Bray kick event rate (blue) vs the Hobbs kick event rate (magenta) for NSNS systems. In green is the LIGO O3 NSNS constraint, and the shaded green region is the uncertainty. We can see that the Bray kick produces likely too high of an event rate density, only being concordant with the upper uncertainty bound.

in Figs. 2.5, 2.2.4, and 2.7 were all generated using the TAKAHE library.

2.3.1 | Definition of compact objects

A final detail we must include within TAKAHE is a way to algorithmically select which binaries are NSNS systems. The canonical constraints on our compact objects, is shown in Table 2.2. This induces a *mass gap* between 2.5 and $5M_\odot$; whilst objects are known to lie within this gap – such as GW190814 (discussed in Abbott et al., 2020b) which has a secondary mass of $2.6M_\odot$.

Type	$M_1(M_\odot)$	$M_2(M_\odot)$
NSNS	< 2.5	< 2.5
BHNS	> 5	< 2.5
BHBH	> 5	> 5

Table 2.2: Canonical constraints on the masses of compact objects. Notice in particular that this induces a mass gap - classically, we expect there to be no compact objects between $2.5M_\odot$ and $5M_\odot$. The existence of GW190814 in Abbott et al. (2020b) implies that the gap is either not as wide (if the object is a Neutron Star) or likely doesn't exist (if it is a Black Hole). We take the maximum mass of a Neutron Star from Rezzolla et al. (2018) (for a nonrotating Neutron Star, and operating conservatively). The minimum mass of a stellar black hole is taken from Farr et al. (2011).

TAKAHE is capable of modifying this mass gap or eliminating it altogether. For instance, we can impose a direct cutoff (i.e., no mass gap) of $2.5M_\odot$. We assume that there is *no* mass gap, and that the NS/BH cutoff is $2.5M_\odot$.

Now that we have established both the theory and prior work, we can move into refining the Bray kick, and computing period-eccentricity distributions.

Chapter 3

Event Rate Densities

There are more things in heaven and earth, Horatio, than are dreamt of in your philosophy.

Hamlet

In the paper originally proposing the Bray kick, Bray writes that "While we have examined one observable test, the model needs to undergo further observational tests to provide stronger evidence of its validity" (Bray and Eldridge, 2018). Moreover, in Fig. 2.7, we showed that the Bray kick currently overestimates the event rate density we expect to see today. Revision of the supernova kick model is critical to providing a way to determine *a posteriori* the properties of the binary system that generated a given gravitational wave signal. Thus, we look to different models of the Bray kick, and aim to narrow down the two parameters α and β which define it.

The way this is performed is iterative. We start by simulating 961 different combinations of α/β parameters, each combination comprising some 19000 models, totalling 18,259,000 models at solar metallicity. Prior work, such as Tang (2019), has identified that for NSNS systems, the impact of metallicity is essentially negligible, and we thus take $Z = Z_{\odot}$.

The coarse grid of α, β parameters that we use takes steps in increments of $\mathrm{d}\alpha = \mathrm{d}\beta = 10$ km/s. α is taken to range from 0 to 300, and β is taken to range from -200 to 100. These were chosen as the output of the Tui code, and future work should take finer steps around the resultant values of α and β to refine their uncertainties.

3.1 | $\beta = 0$: a viable candidate

The simplest case is $\beta = 0$, and varying α. To physically motivate this, consider again eqn(2.3): in the limit as $M_{\mathrm{ejecta}} \to 0$, we may expect that the kick velocity should go to zero, which necessitates that $\beta \simeq 0$. Allowing $\beta \neq 0$, though, does not permit this intuition – in that case, systems with $\frac{M_{\mathrm{ejecta}}}{M_{\mathrm{remnant}}} < -\frac{\beta}{\alpha}$ will have a negative kick velocity. The standard argument motivating the existence of a non-zero β parameter is that any potential negative velocity is sourced by the gravitational tugboat mechanism (see Janka, 2017; Wongwathanarat et al., 2013). This is where an asymmetric mass ejection causes clumps of ejecta mass to effectively "recoil" the compact object. However it is argued (Mandel and Müller, 2020) that this may be unphysical and violate conservation of momentum, as this has the velocity

oriented in the direction in which greatest momentum is carried away. Whilst we do not comment on whether or not $\beta = 0$ is physically required, we consider it nonetheless.

We thus consider the case where $\beta = 0$ and we find that the favored value of α to replicate the LIGO results from Abbott et al. (2020a) is $\alpha = 110 \pm 5$, as shown in Fig. 3.1. We note that this lies within the bounds presented by Bray (2018) – as in Table. 2.1, we have $\alpha_{2D} = 100^{+30}_{-20}$ and $\alpha_{3D} = 60^{+30}_{-10}$.

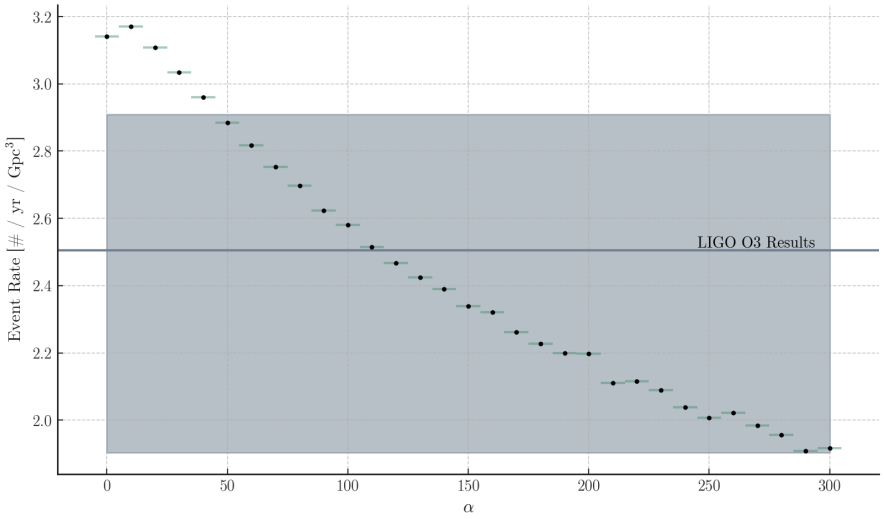


Figure 3.1: The event rate as a function of α for NSNS systems, with $\beta = 0$. We assume Poisonnian uncertainties in the event rate – which are smaller than the pixels, and we assume that the error in α is ± 5 (since the event rates were computed on a grid with step size $h = 10$). The grey box denotes the uncertainty bounds on $\mathcal{R}_{0,\mathrm{LIGO}}$, the event rate density as observed by LIGO.

The uncertainty on α is derived from the coarse nature of the grid, and the desire to have the nominal O3 result encapsulated within our α parameter. However, per Table 1.1, the true observational event rate may be between $\log \mathcal{R}_0 = 2.1$ and $\log \mathcal{R}_0 = 3.1$. Thus, we can widen the uncertainty bound to $\alpha = 110^{+190}_{-60}$. This is sufficient to replicate the nominal LIGO O3 data results, and includes all α values that lie within the LIGO uncertainty band. Additionally, we use Poissonian statistics: the uncertainty in any given value in Fig. 3.1 is $\sigma = 1/\sqrt{N}$ – where N is the number of data points in bin 0 of the event rate histogram for $\alpha = \alpha$ and $\beta = 0$.

Our uncertainty in α is large as a direct result of the wide band of uncertainty for LIGO. Repeating this simulation when more results are released, and the event rate is narrowed down, will prove to be a useful exercise due to the tighter constraints that LIGO will be able to place on their merger rate density.

3.2 | $\beta \neq 0$: Refining the Bray Kick further

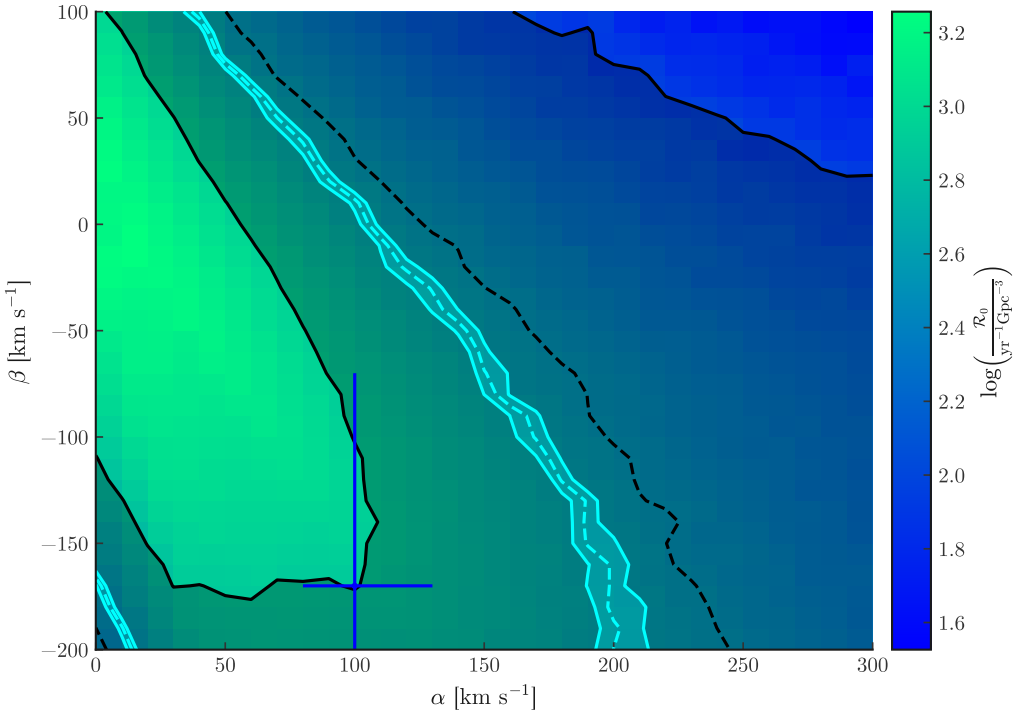


Figure 3.2: α-β plot of the Bray kick event rate. An individual pixel in this plot represents the event rate today, $\log \mathcal{R}_0$, as predicted by Takahe, when the Bray kick is assumed to have the corresponding parameters α and β. The black dashed contour is the observed merger rate density from Abbott et al. (2020a) (pg. 28) and the solid black lines are the corresponding uncertainty values. The cyan dashed contour is the theoretical merger rate density from Eldridge et al. (2018) (Table 1). The fiducial value of the Bray kick, $\alpha = 100, \beta = -170$, is shown as the blue cross.

We now consider the case when $\beta \neq 0$. For this, we iterate over all 961 combinations of α and β, and compute $\log \mathcal{R}_0$ for each combination. Thus, we can generate a plot as in Fig. 3.2, where each cell corresponds to an α and β pair. We overplot different theoretical and empirical results. The cyan line is a contour representing the fiducial value reported in Eldridge et al. (2018), and we can see this is reproducible by a range of parameters. The black line is the result calculated by LIGO from Abbott et al. (2020a). For both the cyan and black lines, we include the uncertainty bands as shaded regions. The blue cross represents the fiducial value of the Bray kick reported in Bray (2018), with associated uncertainty bounds. We note that it does not lie on the Eldridge et al. (2018) contour as, since publication of Eldridge et al. (2018), Tui has been refined regarding the calculation of the rejuvenation age of the binary.

We also note that in Fig. 3.2, there is no single "best fit" to the O3 data results. Rather, there are a number of possible pairings of α and β which would replicate the required results, with β decreasing as α increases. On average, our new kick velocity will likely be higher than the fiducial Bray kick velocity, which is predominantly caused by the highly negative value of β, and decreases the overall strength of the kick. This likely causes more binaries to become unbound after the secondary supernova, and in general widen the binaries, leading to a longer coalescence time.

Fig. 3.2 indicates that the Bray kick is likely to be incorrect. The theoretical predictions from BPASS are shown in that plot by the cyan line. We note that the range of α and β parameters permitted by the O3 data comprises the majority of the parameter space. We concern ourselves with replicating the *nominal* O3 results, leading to 34 considered α, β pairs. If the Bray kick is correct, it will be many further data releases before we have full confidence in our parameters.

To understand whether or not the different models vary beyond having the same $\mathcal{R}_0$, we take three models and compare their DTDs in Fig. 3.3. We select $(\alpha, \beta) = (110, 0)$, $(\alpha, \beta) = (200, -50)$, and $(\alpha, \beta) = (70, 50)$. The first model is chosen as it is our favoured case when $\beta = 0$. The latter two models are chosen as they lie on the LIGO O3 contour, with $(\alpha, \beta) = (70, 50)$ having a lower mean kick velocity over the entire ejecta-remnant mass ratio range, and $(\alpha, \beta) = (200, -50)$ a higher one. There are clear differences between the DTDs – more than just sampling noise. The first and third distributions have "bumps" at approximately 11 Gyr, however this is not present in the second. Moreover, the second plot has no bins with zero height, whereas both the first and third do at around 5 and 6 Gyr. This is exemplified by the resultant merge rate distributions – as the SFR is the same for each, yet the $(110, 0)$ model yields fewer mergers today than the $(70, 50)$ model. Indeed, the $(200, -50)$ model has a small dip in the merge rate centered at 2 Gyr. We conclude that something that looks at the merge rate over a longer time will allow us to determine the kick more clearly.

We now turn to what pair of parameters (α, β) best replicates *both* the nominal LIGO O3 value and validates observed distributions. First, we observe Fig. 3.2, and extract the α and β values that lie along the LIGO O3 contour (specifically, the contour for the nominal value). These form a set of possible pairings in α and β. We will examine these pairings in further detail, by observing their period-eccentricity distributions, in Chapter 4.

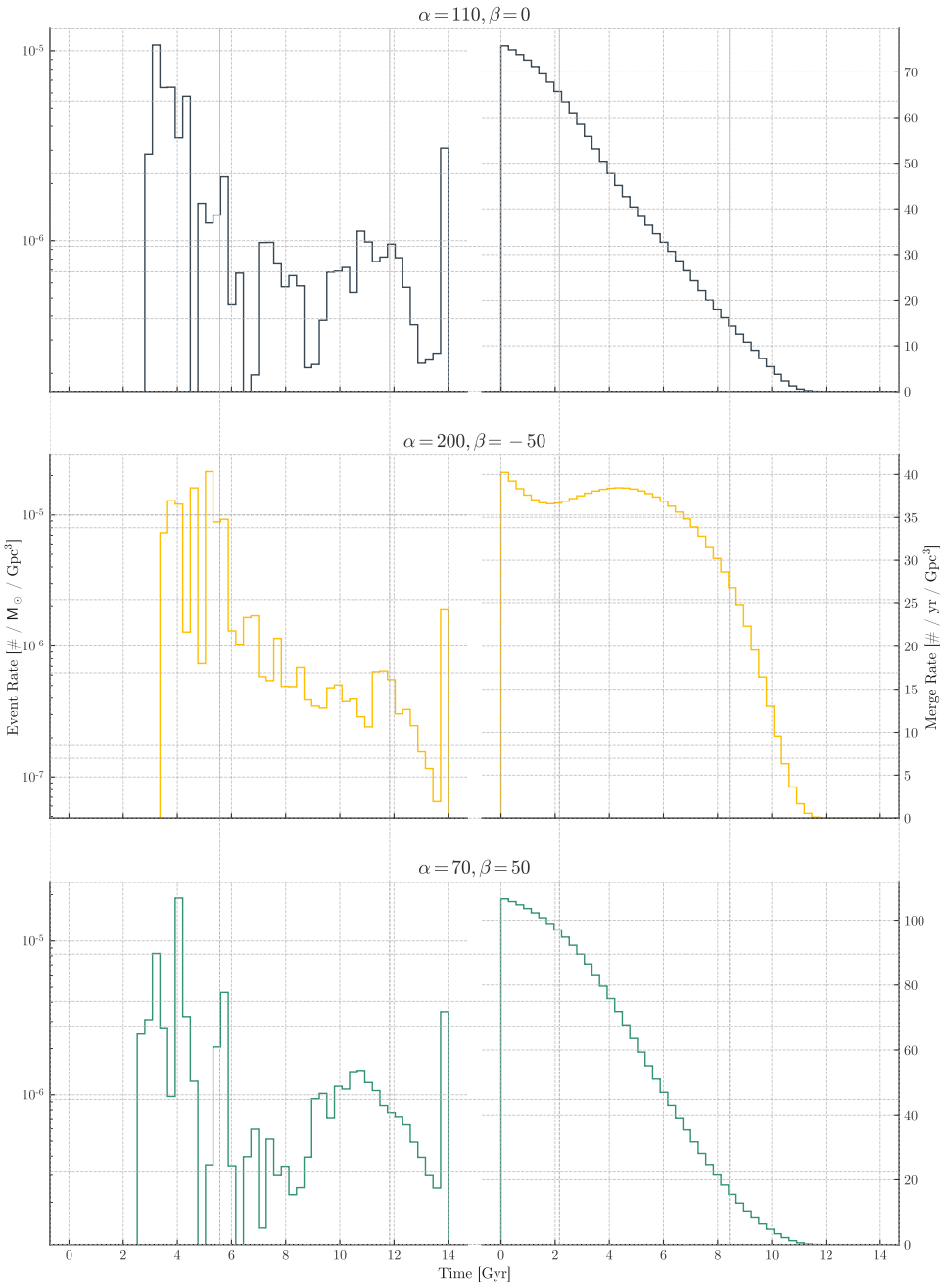


Figure 3.3: The DTD (left) and Merge Rate distribution (right) for various parameters of the Bray kick model. We note that the DTDs shown here resemble those as shown in Fig. 2.5, though this simulation was performed with fewer models than that one. Note also that the merge rate plot is different to Fig. 2.7 as this simulation is only performed at $Z = Z_\odot$, whereas Fig. 2.7 includes all BPASS metallicities.

Chapter 4

Period-Eccentricity distributions of NSNS systems

What's so amazing that keeps us stargazing? And what do we think we might see?
Someday we'll find it, the rainbow connection. The lovers, the dreamers, and me.

Kermit the Frog

In Chapter 3, we examined the event rate for different permutations of the Bray kick parameters. But seeking to understand the case when $\beta \neq 0$ left us with a large array of potential permutations of α and β. Thus, to refine which combination is correct, we now turn to replicating the observed distributions from Table 1.2. This can be shown by way of a period-eccentricity plot. We make the following assumptions:

(1) We take the SFR to be given by

$$\Psi\left(z, \frac{Z}{Z_\odot}\right) = 1\mathrm{M}_\odot/\mathrm{yr} \tag{4.1}$$

as this is the simplest case, and a galactic SFR complicates the calculation.

(2) We place stars on the period-eccentricity plot after the secondary supernova, and after a further period of stellar evolution – in BPASS terms, this is the rejuvenation age + evolution age.

We generate our period-eccentricity plots as follows: we consider a three-dimensional space with axes of time t, period $\log P$, and eccentricity e. The time bins are separated by a constant timestep $\mathrm{d}t = 100$ Myr, the eccentricity bins by $\mathrm{d}e = 10^{-4}$, and the period bins by 0.1 dex. For every star in the sample (at fixed metallicity – we take $Z = Z_\odot$), we begin evolution at the evolution age + the rejuvenation age, and forward propagate until either the merger, or a 10km orbit, or until 10^{11} years have passed (past the age of the Universe). In this way we obtain a curve in time-period-eccentricity space representing the evolution of the binary star system. The code to compute this integration is given in Appendix D. We illustrate this with a plot of a single NSNS system evolving through this space in Fig. 4.1, and the resultant period-eccentricity distribution in Fig. 4.2.

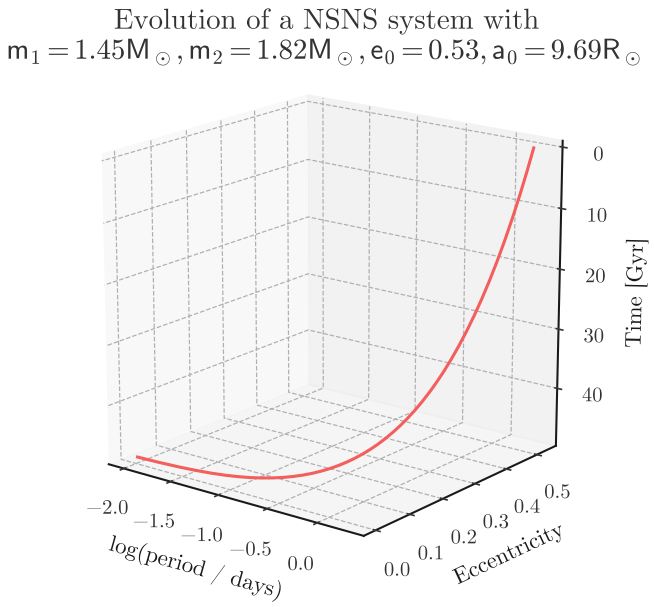


Figure 4.1: The evolution of a single NSNS system, with $m_1 = 1.45 M_\odot, m_2 = 1.82 M_\odot, a_0 = 9.69 R_\odot, e_0 = 0.53$, and a pre-merger evolutionary time of 39.2 Myr. This system occurs in the Bray kick dataset we use. A standard period-eccentricity distribution is obtained by projecting this plot into the period-eccentricity plane, and the result of this is shown in Fig. 4.2.

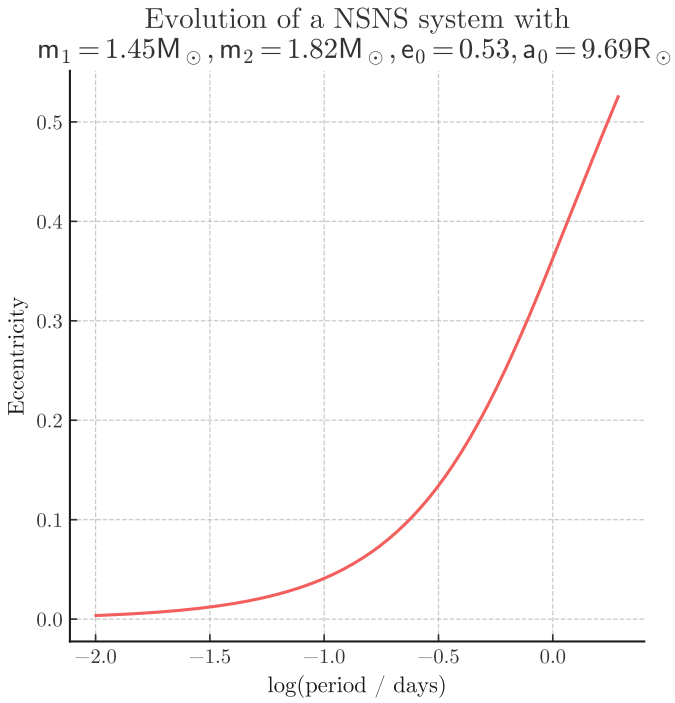


Figure 4.2: The evolution of the NSNS system shown in Fig. 4.1. The plot has been projected into the period-eccentricity plane.

Our code in particular discretises the cube; the cube is sliced into individual chunks separated by $\mathrm{d}t = 10^8$ years. When an individual binary is propagated through time, we take every point in period-eccentricity space, compute the time bin it belongs in, and place the point in that period-eccentricity-time bin. This means that, for ecery slice of constant time, there is a matrix of points in the period-eccentricity plane, which we can sum element-wise to resolve the entire period-eccentricity plot throughout time.

We will consider four kick models in turn: the Hobbs kick, the fiducial Bray kick, the $\beta = 0$ Bray kick, and the $\beta \neq 0$ Bray kick. First, we must determine the optimal parameters for $\beta \neq 0$.

4.1 | Determining the optimal parameters for $\beta \neq 0$

In Chapter 3, we gathered possible α, β parameters that would produce event rate densities that would replicate the LIGO O3 data results. We now seek to determine which of these parameters also replicates observed NSNS systems most accurately. This will be our favoured parameter set for $\beta \neq 0$.

For every pair of α and β, we compute the final period-eccentricity distribution, and evaluate its goodness-of-fit to determine which distribution best models the observed data as given in Table 1.2 on page 6. The period-eccentricity distributions are shown in Fig. 4.4 on page 44. We compute the goodness-of-fit similarly to that outlined in Appendix A of Vigna-Gómez et al. (2018). Crucially, though, we differ in computing $p(\log P, e|M)$ – the probability of observing a system with period $\log P$ and eccentricity e, given the kick model M: they write

$$p(\log P, e|M) = \sum_{j=1}^{n_{\text{binaries}}} \sum_{k=1}^{n_{\text{timesteps},j}} \frac{\Delta t_k}{T} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{4.2}$$

whereas we write

$$p(\log P, e|M) = \sum_{j=1}^{n_{\text{binaries}}} w_j N(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j) \tag{4.3}$$

where w_j is the content of the bin corresponding to (P_j, e_j), and $N(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$ is the 2D Gaussian function with mean $\boldsymbol{\mu}_j = (\log P_j, e_j)^\top$ and covariance

$$\boldsymbol{\Sigma}_j = \begin{pmatrix} b^2_{\log P_j} & 0 \\ 0 & b^2_{e_j} \end{pmatrix}$$

Thus, instead of weighting the binaries by the time they spend in each bin, we weight each by the relative amount of "stellar activity" in each bin. Similarly to Vigna-Gómez et al. (2018), we use the "rule of thumb" bandwidth parameters from Silverman (1986), such that

$$b_{\log P} = \frac{0.9 m_P}{n^{1/5}} \quad \text{where} \quad m_P = \min\left\{\sqrt{\mathrm{Var}(\log P)}, \frac{\mathrm{IQR}(\log P)}{1.349}\right\} \tag{4.4}$$

$$b_e = \frac{0.9 m_e}{n^{1/5}} \quad \text{where} \quad m_e = \min\left\{\sqrt{\mathrm{Var}(e)}, \frac{\mathrm{IQR}(e)}{1.349}\right\} \tag{4.5}$$

where $\mathrm{IQR}(A)$ means the inter-quartile range of parameter A. This approach assumes a Gaussian distribution of observed NSNS systems: earlier we assumed Poissonian statistics, here we use Gaussian statistics. This is because the Central Limit Theorem dictates that in the limit as the number of datapoints becomes large, a Poisson distribution may be approximated by a Gaussian one.

Formally, when we compute $p(\log P, e|M)$, we are computing the log-likelihood, $\log \mathcal{L}$. The values of $\log \mathcal{L}$ we compute are given in Table 4.1. Moreover, we find that the four models we compute have log-likelihoods of 482±13 ($\beta = 0$), 1002 ±13 ($\beta \neq 0$), 1112±13 (fiducial Bray), and 179±13 (Hobbs). That is, the fiducial Bray kick is best at replicating observed distributions, whereas the Hobbs kick is the worst. We must, however, remember that this is only one metric of determining the "best kick". The observable we want to maximise determines which kick is "best" of the four considered.

It is also pertinent to note that this Kernel Density Estimation (KDE) is sensitive to the chosen bin size. We select bins linearly spaced in eccentricity with a width of 0.01, and linearly spaced bins in $\log P$ with a width of 0.1 dex.

We find that the favored parameters for the Bray kick are therefore $\alpha = 40, \beta = 100$. This particular period-eccentricity distribution is shown as Fig. 4.3.

α	β	$\log \mathcal{L}$	α	β	$\log \mathcal{L}$
40	100	1002±13	150	−50	523±13
40	90	885±13	150	−60	512±13
50	80	749±13	160	−70	460±13
50	70	734±13	160	−80	399±13
60	70	660±13	170	−90	421±13
60	60	571±13	170	−100	480±13
70	50	518±13	180	−110	454±13
80	40	520±13	180	−120	444±13
80	30	527±13	190	−130	429±13
90	20	527±13	190	−140	504±13
100	10	504±13	200	−150	445±13
110	0	482±13	210	−160	526±13
110	−10	409±13	210	−170	543±13
120	−20	425±13	200	−170	521±13
130	−20	330±13	210	−180	542±13
130	−30	486±13	220	−190	560±13
140	−40	472±13	220	−200	607±13

Table 4.1: The log-likelihoods of the models given in Fig. 4.4. The maximum $\log \mathcal{L}$ is given by the model $\alpha = 40, \beta = 100$, which is therefore the favoured model we select.

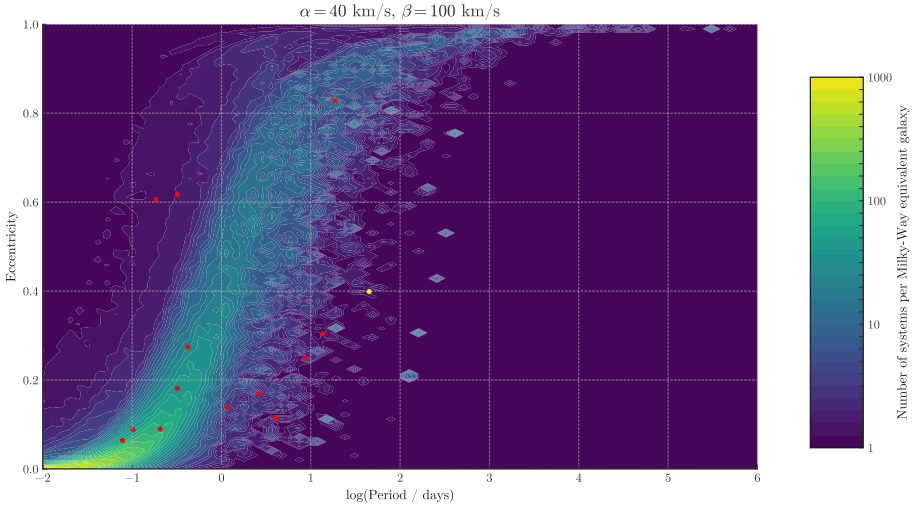


Figure 4.3: The period-eccentricity distribution for $\alpha = 40, \beta = 100$. Overplotted are the observed pulsars from 1.2, and J1930-1852 from Swiggum et al. (2015) is highlighted in yellow.

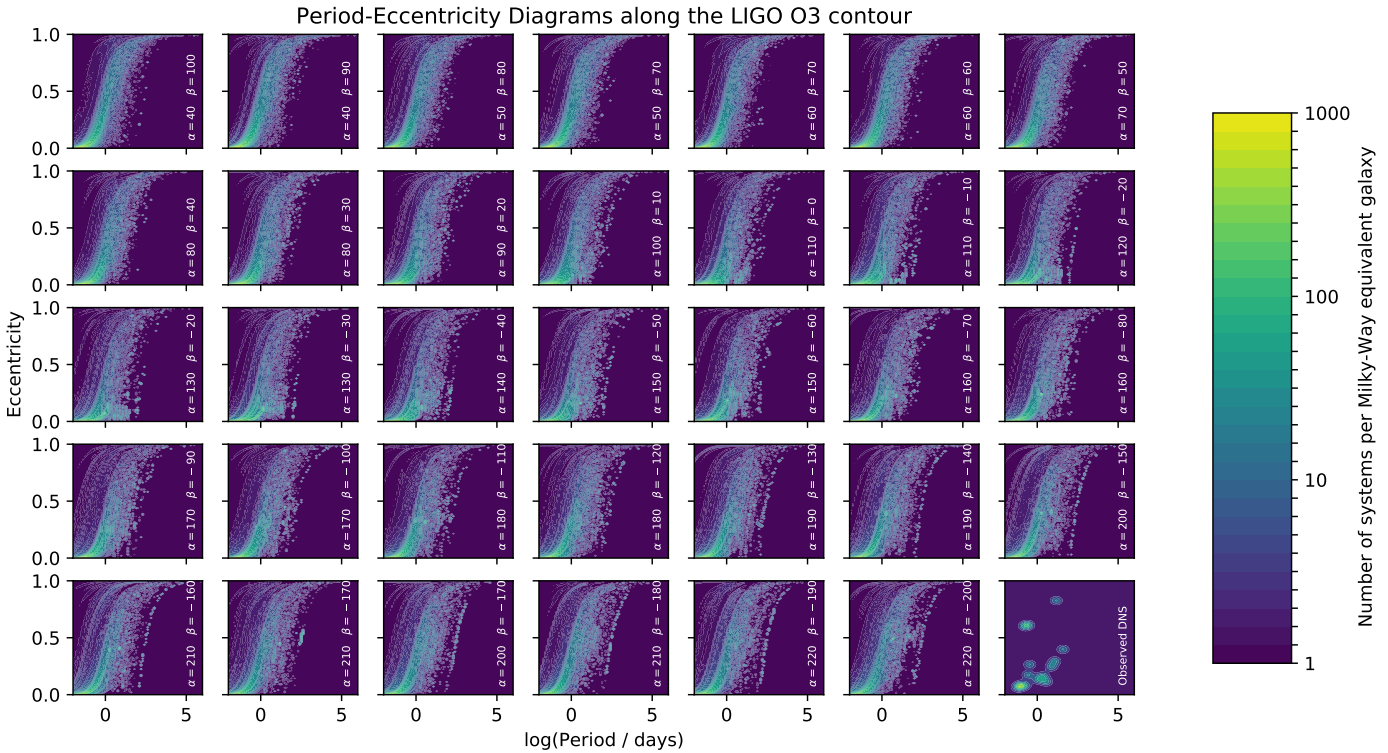


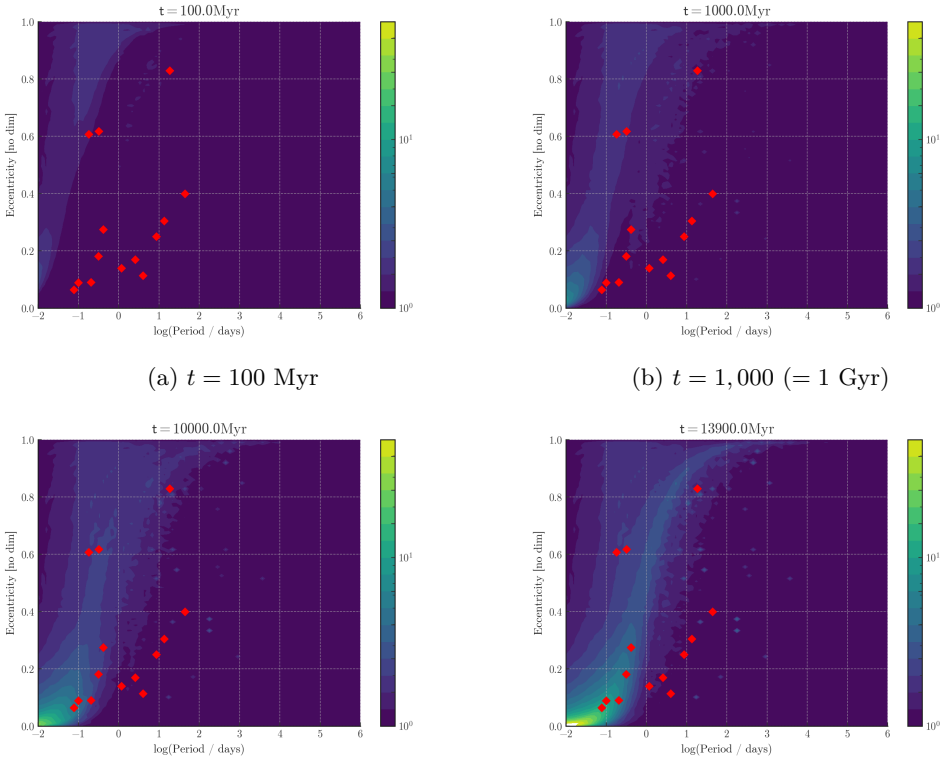
Figure 4.4: The period-eccentricity distributions for the systems along the LIGO O3 contour. The x-axis of each plot is $\log P$ (ranging from -2 to 6), and the y-axis is eccentricity (ranging from 0 to 1). The colour axis is assumed to have a SFR of 1 $\mathrm{M}_\odot/\mathrm{yr}$, and is culmulative throughout time. The "Observed DNS" panel has a Gaussian blur applied with $\sigma = 2$, to illustrate the intrinsic uncertainty in measurement. The best-fit α, β parameters are shown, blown-up, as Fig. 4.3.

4.2 | Results

The following plots are *cumulative*, and show the total evolution throughout time. The z-axis the total number of systems with that period and eccentricity, in units of yr^{-1} Gpc^3.

4.2.1 | The Hobbs kick

First, we consider the Hobbs kick, as the baseline kick to compare against. We find that Hobbs is unable to replicate some pulsars, including J1930-1852 from Swiggum et al. (2015) ($\log P = 1.653791, e = 0.399$). Replication of this pulsar is critical to the validation of any kick model; this is because it lies at the fringe of known parameters. Thus, we can see that the Hobbs kick is likely incomplete, as it does not have high period ($\log P \gtrsim 10$ days) systems at low eccentricity.



(a) $t = 100$ Myr

(b) $t = 1,000$ ($= 1$ Gyr)

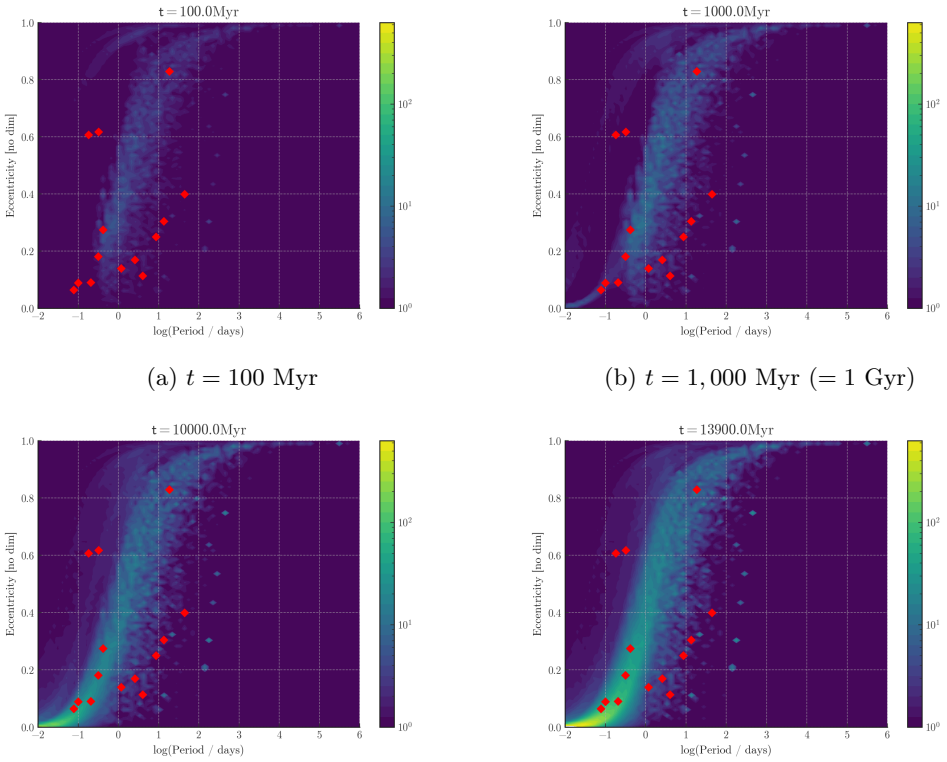
(c) $t = 10,000$ Myr ($= 10$ Gyr)

(d) $t = 13,900$ Myr ($= t_{\mathrm{H}}$).

Figure 4.5: The Period-Eccentricity distributions for the Hobbs Kick. This proceeds in 100 Myr steps.

4.2.2 | The fiducial Bray kick

The fiducial Bray kick takes $\alpha = 100, \beta = -170$. Similar to the Hobbs kick, we see that certain pulsars such as J1930-1852 do not appear to be replicated until later stages of evolution. We begin to see a "zone of avoidance" at earlier times in the Universe. This is an area of low density, which in Fig. 4.6b is roughly a triangle bounded by the coordinates $(0.2, 0.9)$, $(-0.9, 0.6)$, and $(0, 0.6)$.



(a) $t = 100$ Myr

(b) $t = 1,000$ Myr $(= 1$ Gyr$)$

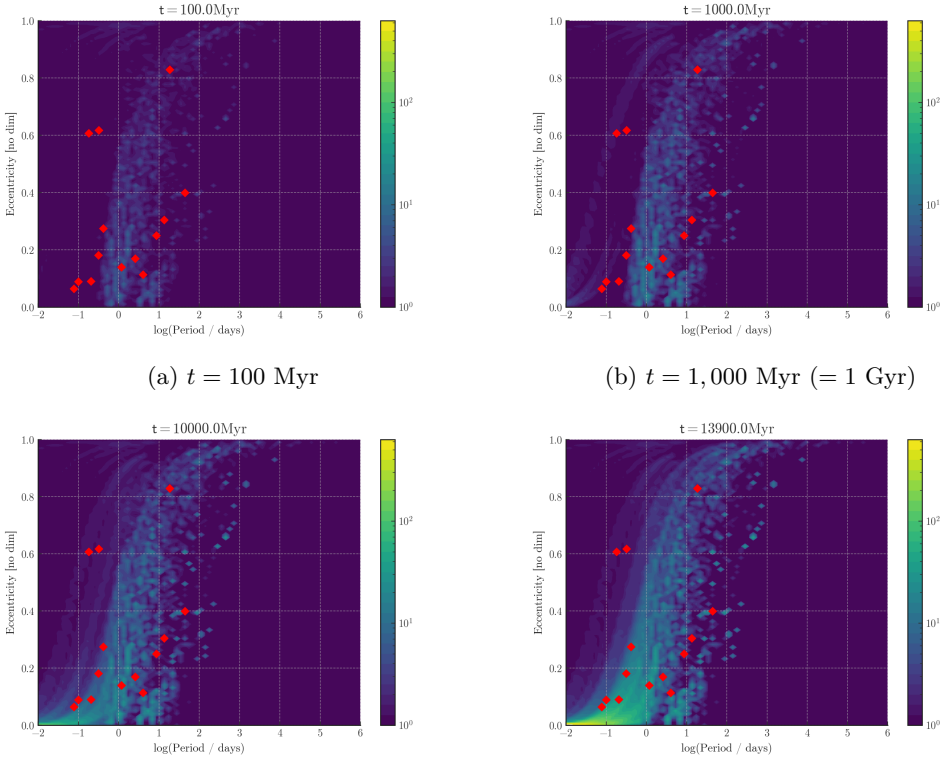
(c) $t = 10,000$ Myr $(= 10$ Gyr$)$

(d) $t = 13,900$ Myr $(= t_{\mathrm{H}})$.

Figure 4.6: The Period-Eccentricity distributions for the fiducial value of the Bray Kick. This proceeds in 100 Myr steps.

4.2.3 | The case when $\beta = 0$

We take the favoured $\beta = 0$ value from Chapter 3, which is $\alpha = 110, \beta = 0$. This results in a clearer zone of avoidance, which spans down to approximately $(-1.5, 0.2)$ and $(-0.5, 0.2)$. We can see that this possibly admits J1930-1852, although at times very close to the Hubble time.



(a) $t = 100$ Myr

(b) $t = 1,000$ Myr $(= 1$ Gyr$)$

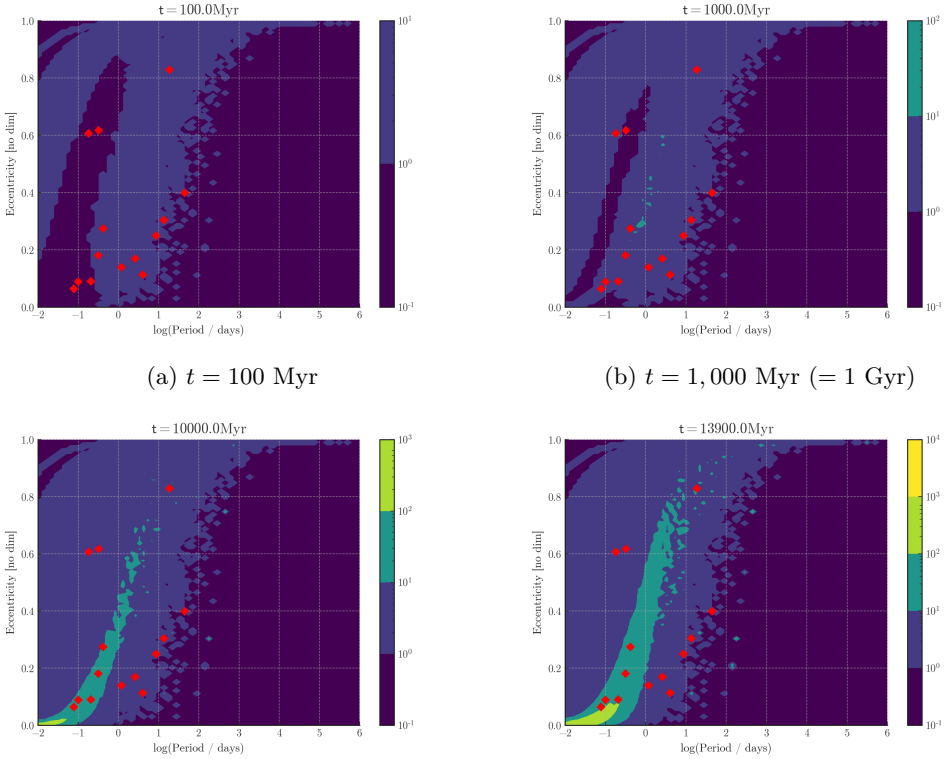
(c) $t = 10,000$ Myr $(= 10$ Gyr$)$

(d) $t = 13,900$ Myr $(= t_{\mathrm{H}})$.

Figure 4.7: The Period-Eccentricity distributions for the Bray Kick with $\alpha = 110, \beta = 0$.. This proceeds in 100 Myr steps.

4.2.4 | The case when $\beta \neq 0$

The broadest case we consider is $\beta \neq 0$, which corresponds to the favoured parameters $\alpha = 40, \beta = 100$. This presents the clearest view of the zone of avoidance, which we can now see is a region that does not have any systems permitted at earlier times, but closes soon after 1 Gyr.



(a) $t = 100$ Myr

(b) $t = 1,000$ Myr ($= 1$ Gyr)

(c) $t = 10,000$ Myr ($= 10$ Gyr)

(d) $t = 13,900$ Myr ($= t_{\mathrm{H}}$).

Figure 4.8: The Period-Eccentricity distributions for the Bray Kick with $\alpha = 40, \beta = 100$. This proceeds in 100 Myr steps.

4.2.5 | Discussion

The four period-eccentricity distributions can also be visualised as GIFs. The GIFs themselves are visible on GitHub[1] and selected frames are shown in the sections above.

We find that the Hobbs model produces fewer NSNS systems of mid-eccentricity and higher periods (on the order of 50-100 days). Notably, this includes the pulsar J1930-1852 from Swiggum et al. (2015), which has an orbital period of ∼45 days and an eccentricity of 0.399. In their work, they write that "its unique spin and orbital parameters challenge models that describe DNS formation". Consequently, admissibility of this particular pulsar is a critical test of any population synthesis model. The fact, then, that the Bray kick admits it whilst the Hobbs kick only barely does provides further evidence away from the Hobbs kick.

[1] The filename convention is `evolving_cube_{kick}_{alpha}_{beta}.gif`. For the purposes of this convention, the Hobbs kick has `{alpha}=0` and `{beta}=0`. https://github.com/UoA-Stars-And-Supernovae/Richards-thesis

4.3 | Highly eccentric systems

Some NSNS systems have merger timescales orders of magnitude higher than the age of the Universe (as seen in Fig. 4.9). Thus, we may never be able to experimentally observe their mergers. However, we will soon (with the launch of LISA) be able to observe some of these binaries *before* they merge. We now look at the distribution of such highly eccentric systems today. We first integrate past the end of the Universe, to a maximum cutoff time of 10^{11} years. We consider all 961 combinations of the varied Bray kick parameters, as in Chapter 3, and we only consider systems who have final eccentricities after 10^{11} years of > 0.8. The precise code to accomplish this is given as Appendix E. As a note, we take $Z = Z_{\odot}$ - prior work such as Bray and Eldridge (2018) has shown that metallicity has a negligible impact on the kick for NSNS systems, and is only prevalent for BHBH systems.

Taken together, Figs. 4.9 and 4.10 show that models that have a high frequency of highly eccentric mergers also tend to have longer average merger timescales. These mergers, then, depend on star formation nearer the beginning of the Universe. Conversely, models that have a lower frequency of highly eccentric mergers depend on more recent star formation activity. A future way of limiting potential kick parameters will therefore be comparing the DTD of host galaxies to observed distributions, and looking at the merger rate as a function of redshift.

Fig. 4.10 also shows the percentage of highly eccentric systems we expect to see today, relative to the total number of NSNS systems generated from that α, β pair, as a function of the Bray kick parameters α, β. Although there are no quantifiable sources as to the exact rate we should expect to see today, we intuitively expect the rate to be low. Two of the known binary neutron stars, for example, have eccentricities of 0.6 (the Hulse-Taylor binary, see Weisberg et al. 1981) and 0.09 (PSR J0737-3039B, see Noutsos et al. 2020). We may never have experimental confirmation of the exact percentage of the current NSNS systems that have high eccentricity – highly eccentric binaries tend to have merger timescales several orders of magnitude above the Hubble time – so we conservatively estimate it at an upper bound of 5%.

This threshold admits 50 possible permutations of the Bray kick parameters. In particular, neither of the kicks so far considered are in this sample, though a slight relaxation of the threshold to 9% admits the $\alpha = 110, \beta = 0$ solution. We can, however, discern that the magnitude of the kick does have an appreciable impact on the eccentric population we observe today – contrast, for example, $\alpha = 140, \beta = 90$ in Fig. 4.11 against $\alpha = 50, \beta = -140$ in Fig. 4.12. The first kick has a higher magnitude for all ejecta-remnant ratios, and has a mean frequency across β of 9.9%. The second leads to a higher mean merger timescale – as seen in Fig. 4.9 – and has a mean frequency across β of 18.1%. One concludes that a high magnitude kick produces events with higher initial eccentricities, and therefore higher merger timescales.

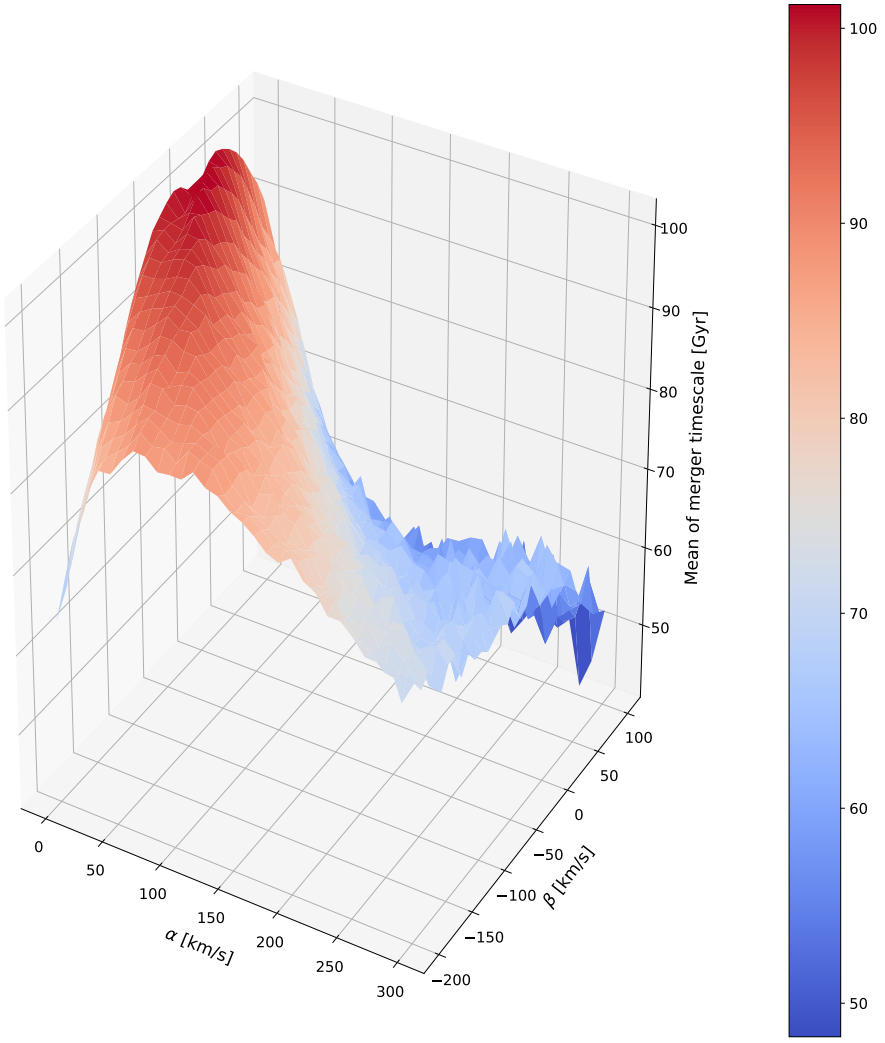


Figure 4.9: The average merger timescale for NSNS systems. The z-value corresponding to a particular α, β pair is computed by integrating a populations of NSNS systems from formation until coalescence, with cutoffs at $a = 10$ km (the presumption that they merge), $t = 10^{11}$ years (past the age of the Universe), or $e \notin [0, 1]$ (physical impossibility). Therefore, the maximum value any particular cell can take is 100 Gyr, which indicates the vast majority of these systems merge at longer than the age of the Universe.

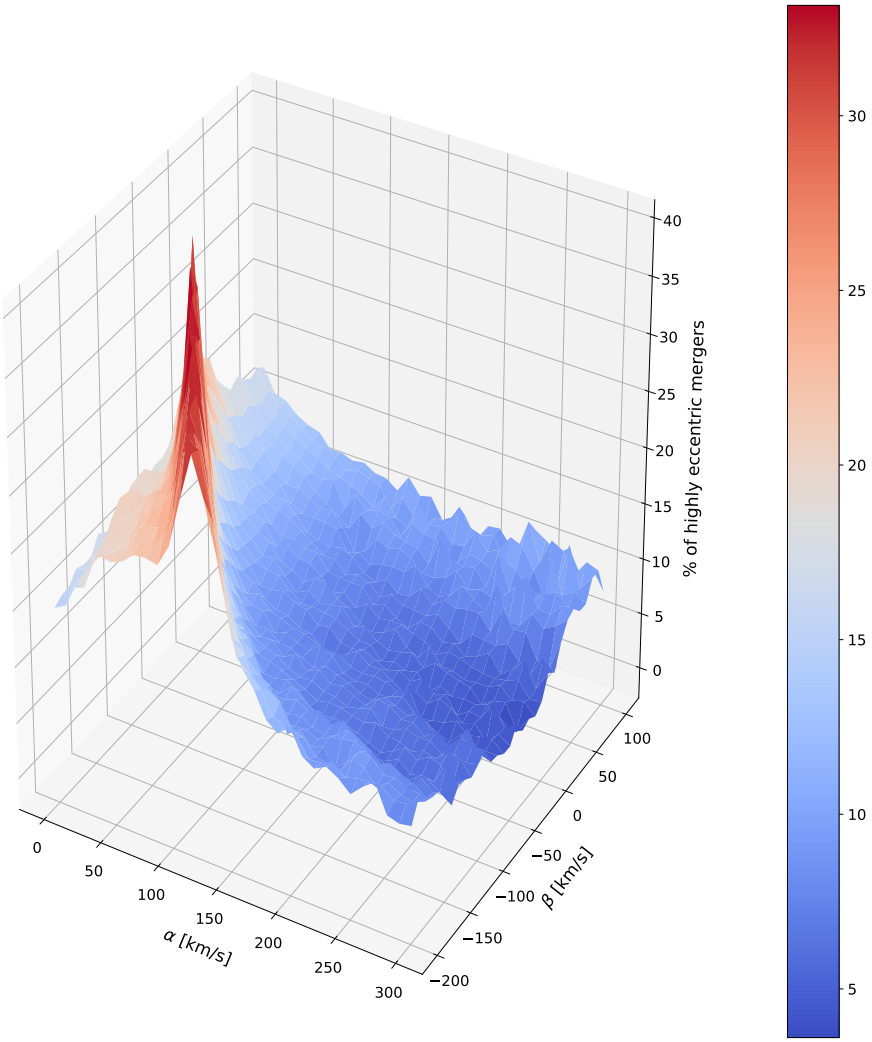


Figure 4.10: The normalised frequency of highly eccentric ($e_{\text{final}} > 0.8$) mergers, as a function of kick magnitude. Individual α, β parameters have the contents of their bins normalised to the number of NSNS systems in that sample.

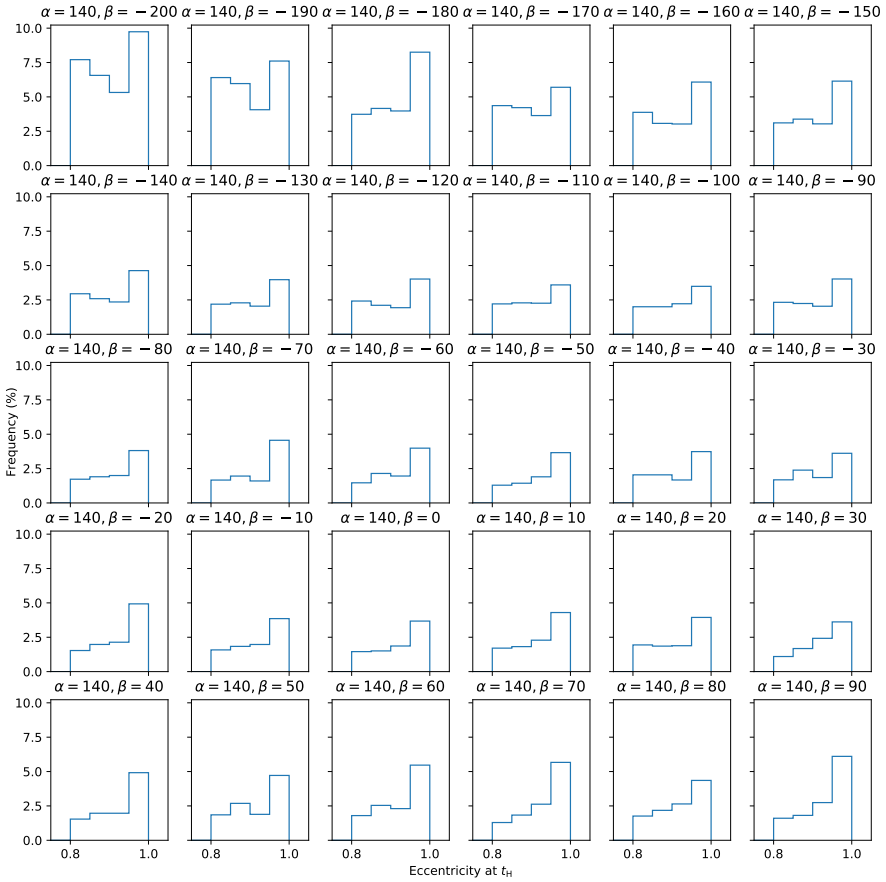


Figure 4.11: The current-time eccentricity distributions for $\alpha = 140$ km/s.

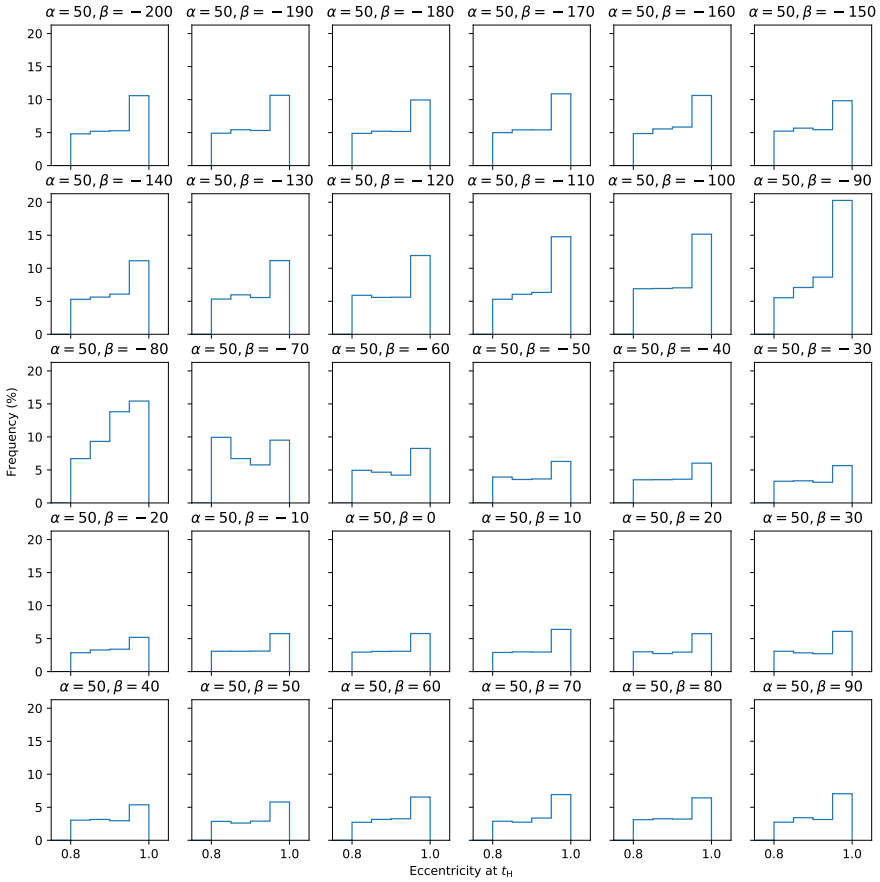


Figure 4.12: The current-time eccentricity distributions for $\alpha = 50$ km/s.

4.4 | Conclusions

The primary conclusion that we can draw from this approach is that both the form and magnitude of the kick has an appreciable effect on the nature of the Period-Eccentricity distribution. This approach is how we compare the observed DNS systems to the modelled ones; as in Section 3.1, we can run a KDE (assuming a Gaussian prior), and thereby compute the goodness-of-fit. Questions remain around the precise nature of the natal supernova kick, though, which we will outline in our concluding remarks in Chapter 5.

Chapter 5

Conclusion

I'll be a story in your head. But that's OK: We're all stories, in the end. Just make it a good one, eh? Because it was, you know, it was the best: A daft old man, who stole a magic box and ran away.

The Eleventh Doctor

In Chapter 1, we outlined two overarching themes to this thesis.

(1) Refining the Bray kick model, to more accurately predict the expected merger rate of binary stars.

(2) Efficiently and accurately generating Period-Eccentricity distribution diagrams for these stars, and observing the influence of the supernova kick on them.

The first led us to refining the Bray kick: we found that no one value replicated every parameter we compared to. We followed the arguments of Mandel and Müller (2020) and required $\beta = 0$, and found that while this resolved questions surrounding conservation of momentum, and matched the current estimated value of $\mathcal{R}_0$, it did not fully resolve the question of how many highly eccentric binaries we should observe. Similarly, when we required $\beta \neq 0$, we found a value of α and β that replicated the LIGO observations, but had a high value of extant highly eccentric NSNS binaries. We concluded that the Bray kick, if it is correct, must be incomplete, and conjectured that a modified Maxwell-Boltzmann distribution may resolve this problem. This modified distribution would have the distribution parameter σ replaced by the kick velocity $\alpha\left(\frac{m_{\rm ej}}{m_{\rm rem}}\right) + \beta$. A complete summary of results for our analysis are presented in Table 5.1.

The second question led us to generating period-eccentricity distributions for NSNS systems under various kick models. We found that both the form and magnitude of the kick has an appreciable impact on the nature of the distribution. Namely, we found that the Hobbs kick produced fewer mid-eccentricity systems with periods on the order of 50-100 days, such as the pulsar J1930-1852. The criticality of a population synthesis model including this particular pulsar provided indication that the Hobbs kick is likely not complete.

Yet, in this work, we ultimately considered just two kicks: the Bray kick and the Hobbs kick. More kick prescriptions exist; for instance the Baker-Boggs kick (Baker et al., 2008), which takes the kick velocity to be related to the mass ratio $q = \frac{M_1}{M_2}$ as

Model Name	α km/s	β km/s	$\log \mathcal{R}_0$ # / yr / Gpc^3	Frequency of eccentric current NSNS systems*	Mean Merger Timescale Gyr	$\log \mathcal{L}$
Bray ($\beta = 0$)	110	0	2.5140 ± 0.0002*	8.37%	71.8	482±13
Bray ($\beta \neq 0$)	40	100	2.526 ± 0.009*	13.26%	69.3	1002 ±13
Bray Fiducial	100	-170	2.852 ± 0.004*	33.4%	89.9	1112±13
Hobbs	-	-	2.28 ± 0.05	11.70%	51	179±13

Table 5.1: A summary of key findings from this work. The uncertainties in $\mathcal{R}_0$ are Poissonian ($1/\sqrt{N}$, where N is the number of data points in the $z = 0$ bin).
*(at $Z = Z_\odot$)

$$v_{\rm kick} \propto \left(\frac{q}{(1+q)^2} \right)^3$$

As we have shown that the form of the kick has an appreciable impact on the period-eccentricity distributions, further work should be undertaken to determine, of more models, which is most accurate to the LIGO merger rate densities predicted.

Further work may include refining the goodness-of-fit estimation to more closely replicate Vigna-Gómez et al. (2018). This would allow us to more properly refine the period-eccentricity distributions.

If the Bray kick is incomplete, we may consider alternative sampling mechanisms. For instance, the Hobbs kick has a free parameter σ which Hobbs et al. (2005) takes to be constant at $\sigma = 265$ km/s. We may use the linear Bray kick sampling mechanism to instead take $\sigma = \alpha\left(\frac{M_{\rm ejecta}}{M_{\rm remnant}}\right)$. This further revision of the kick will be undertaken in future work, and the primary objective of this revision will be to adopt a Maxwell-Boltzmann formalism for sampling kick velocity. Briefly, whilst before we sampled linearly from the distribution given in eqn(2.3), we will sample from a Maxwell-Boltzmann distribution, which has a probability distribution function given by

$$P(v)\,\mathrm{d}v = \sqrt{\frac{2}{\pi}}\frac{v^2}{\sigma^3}\exp\left(-\frac{v^2}{2\sigma^2}\right)\mathrm{d}v \tag{5.1}$$

thus adapting the Hobbs kick. A further point of analysis we can perform is investigating the period-eccentricity distributions for the different kick models.

Stargazing is a uniquely human endeavour, and the urge to explore has pioneered a new age in astronomy. The coming decades will certainly be a thrilling time for gravitational wave astronomers.

Appendices

Appendix A

Derivation of Kepler's Three Laws

Binary star orbits follow a modified version of Kepler's laws of planetary motion. We briefly state the three laws of Kepler:

(1) All planetary orbits are elliptical with the sun at one focus.

(2) A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.

(3) The square of the period of the orbit is proportional to the cube of the semimajor axis (i.e., $T^2 \propto a^3$).

The relevant modifications to Kepler's laws such that they apply to binary stars too are:

(1) All binary stars orbit the barycentre in an ellipse with the barycentre as a focus.

(2) A line segment joining the two stars sweeps out equal areas during equal intervals of time.

(3) The square of the period of each star is proportional to the cube of the semimajor axis (i.e., $T^2 \propto a^3$).

Kepler's laws may be derived[1] as follows.

A.1 | Derivation of the First Law

First we define the vector $\mathbf{r}$ such that

$$\mathbf{r} = \mathbf{r}_\mathrm{b} - \mathbf{r}_\mathrm{a} \tag{A.1}$$

[1] As a historical note, the third law was derived empirically by Tycho Brahe and proved formally with Newtonian mechanics.

with the reduced mass

$$\mu = \frac{M_\mathrm{a} M_\mathrm{b}}{M_\mathrm{a} + M_\mathrm{b}} \tag{A.2}$$

and total mass

$$M = M_\mathrm{a} + M_\mathrm{b} \tag{A.3}$$

Since the angular momentum vector is given by

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \tag{A.4}$$

Its derivative is

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \times \mathbf{p} + \mathbf{r} \times \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = 0 \tag{A.5}$$

where the equality follows as $\mathbf{v} \equiv \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$ is parallel to $\mathbf{p}$, and $\mathbf{r}$ is parallel to $\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} \equiv \mathbf{F}$.

Now we can transform this problem into a one-body problem by considering an object with mass $\mu = \frac{M_\mathrm{a} M_\mathrm{b}}{M_\mathrm{a}+M_\mathrm{b}}$ orbiting a central mass of mass $M_\mathrm{a} + M_\mathrm{b}$. The angular momentum is

$$\begin{aligned}
\mathbf{L} &= \mu \mathbf{r} \times \mathbf{v} && \text{(A.6)} \\
&= \mu r \hat{\mathbf{r}} \times \frac{\mathrm{d}}{\mathrm{d}t}(r\hat{\mathbf{r}}) \\
&= \mu r \hat{\mathbf{r}} \times \left(\frac{\mathrm{d}r}{\mathrm{d}t}\hat{\mathbf{r}} + r\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} \right) \\
&= -\frac{\mathrm{d}r}{\mathrm{d}t}\hat{\mathbf{r}} \times \mu r \hat{\mathbf{r}} - r\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} \times \mu r \hat{\mathbf{r}} \\
&= -\frac{\mathrm{d}r}{\mathrm{d}t}\mu r \underbrace{(\hat{\mathbf{r}} \times \hat{\mathbf{r}})}_{=0} - r^2 \mu \left(\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} \times \hat{\mathbf{r}} \right) \\
&= -r^2 \mu \left(\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} \times \hat{\mathbf{r}} \right) \\
&= r^2 \mu \left(\hat{\mathbf{r}} \times \frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} \right) \\
&= \mu r^2 \hat{\mathbf{r}} \times \frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} && \text{(A.7)}
\end{aligned}$$

We seek an expression for r to determine the trajectory of the stars. Consider, then

$$\begin{aligned}
\mathbf{a} \times \mathbf{L} &= -\frac{GM}{r^2}\hat{\mathbf{r}} \times \left(\mu r^2 \hat{\mathbf{r}} \times \frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t}\right) \\
&= GM\mu\hat{\mathbf{r}} \times \left(\hat{\mathbf{r}} \times \frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t}\right) \\
&= -GM\mu\left[\left(\hat{\mathbf{r}} \cdot \frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t}\right)\hat{\mathbf{r}} - (\hat{\mathbf{r}} \cdot \hat{\mathbf{r}})\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t}\right] \\
&= GM\mu\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t}
\end{aligned}$$

Now consider

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{v} \times \mathbf{L}) &= \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \times \mathbf{L} + \mathbf{v} \times \frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} \\
&= \mathbf{a} \times \mathbf{L} \\
&= \frac{\mathrm{d}}{\mathrm{d}t}(GM\mu\hat{\mathbf{r}}) \qquad \text{(A.8)}
\end{aligned}$$

where $\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = 0$ by eqn(A.5). Now we can perform this integral to yield

$$\mathbf{v} \times \mathbf{L} = GM\mu\hat{\mathbf{r}} + \mathbf{D} \qquad \text{(A.9)}$$

We presume that $\mathbf{D}$ exists wholly in the plane of the orbit, since the other vector quantities do too. We can then introduce the vector $\mathbf{r}$ into this equation:

$$\begin{aligned}
\mathbf{r} \cdot (\mathbf{v} \times \mathbf{L}) &= GM\mu r\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} + \mathbf{r} \cdot \mathbf{D} \\
(\mathbf{r} \times \mathbf{v}) \cdot L &= GM\mu r + rD\cos\theta \qquad \text{(A.10)}
\end{aligned}$$

Now by eqn(A.7) this is simply

$$\frac{L^2}{\mu} = GM\mu r\left(1 + \frac{D\cos\theta}{GM\mu}\right) \qquad \text{(A.11)}$$

We define the eccentricity of the ellipse as

$$e = \frac{D}{GM\mu} \qquad \text{(A.12)}$$

and hence we have

$$r = \frac{L^2/\mu^2}{GM(1 + e\cos\theta)} \qquad \text{(A.13)}$$

which is the equation defining an ellipse, provided that[2] $e < 1$ – thus proving the First Law. As a corollary, since the general equation for an ellipse with semimajor axis a is

$$r = \frac{a(1-e^2)}{1+e\cos\theta} \quad \text{for } 0 \leq e < 1 \tag{A.14}$$

It is immediately apparent that we can write

$$L = \mu\sqrt{GMa(1-e^2)} \tag{A.15}$$

and since L is constant by eqn(A.5), presuming total mass M remains constant we have that μ is constant as well.

A.2 | Derivation of the Second Law

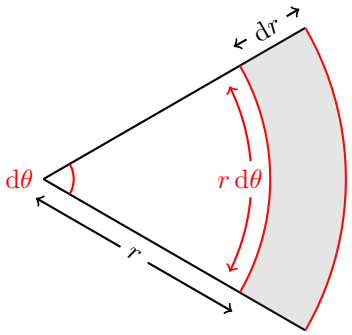


Figure A.1: An infinitesimal area segment of an orbital trajectory. Given the definitions in the figure, we have the area of the shaded segment is $\mathrm{d}A = \mathrm{d}r\,(r\,\mathrm{d}\theta) = r\,\mathrm{d}r\,\mathrm{d}\theta$.

Consider the area of an elliptical orbit (see the shaded area in Fig. A.1). We have

$$\mathrm{d}A = \mathrm{d}r\,(r\,\mathrm{d}\theta) = r\,\mathrm{d}r\,\mathrm{d}\theta \tag{A.16}$$

and integrating over r gives

$$\mathrm{d}A = \frac{1}{2}r^2\,\mathrm{d}\theta \tag{A.17}$$

thus

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}r^2\frac{\mathrm{d}\theta}{\mathrm{d}t} \tag{A.18}$$

[2]Or equivalently that $D < GM\mu$, which we are free to select as required as D is arbitrary.

We can write the velocity of the star in the plane of the ellipse as

$$\mathbf{v} = \mathbf{v}_{\mathrm{r}} + \mathbf{v}_{\theta} = \frac{\mathrm{d}r}{\mathrm{d}t}\hat{\mathbf{r}} + r\frac{\mathrm{d}\theta}{\mathrm{d}t}\hat{\theta} \tag{A.19}$$

Importantly we have

$$v_{\theta} = r\frac{\mathrm{d}\theta}{\mathrm{d}t} \tag{A.20}$$

such that

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}rv_{\theta} \tag{A.21}$$

Since r and v_{θ} are orthogonal, we have

$$rv_{\theta} = \underbrace{|\mathbf{r} \times \theta| = \left|\frac{\mathbf{L}}{\mu}\right|}_{\text{by eqn(A.6)}} = \frac{L}{\mu} \tag{A.22}$$

hence we have

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}\frac{L}{\mu} \tag{A.23}$$

By eqn(A.5), L is constant (and by eqn(A.15), μ is constant as well), hence we have that the rate of change of A is constant over the orbit, and Kepler's second law is established.

A.3 | Derivation of the Third Law

Let the period of the binary star system be given by T. We start by writing

$$A = \int_0^T \frac{\mathrm{d}A}{\mathrm{d}t}\,\mathrm{d}t = \frac{1}{2}\frac{L}{\mu}\int_0^T \mathrm{d}t = \frac{1}{2}\frac{L}{\mu}T \tag{A.24}$$

Immediately establishing that

$$\pi ab = \frac{1}{2}\frac{L}{\mu}T \iff T^2 = \frac{4\pi^2 a^2 b^2 \mu^2}{L^2} \tag{A.25}$$

Further, from the geometry of an ellipse we have

$$b^2 = a^2\left(1 - e^2\right) \tag{A.26}$$

Using this and eqn(A.15) in tandem, we have

$$T^2 = \frac{4\pi^2}{GM}a^3 \tag{A.27}$$

which proves Kepler's third law.

Appendix B

Takahe Event Rate Code

We present the following event rate code that is used in Takahe. Complete code is available on the GitHub repository (https://github.com/Krytic/Takahe).

```
import numpy as np
import pandas as pd
from scipy.special import gamma, gammainc
import takahe
from tqdm import tqdm

def single_event_rate(df, Z, extra_lt=None):
    """Computes the event rate of a single metallicity.

    Computes the event rate for metallicity Z, using Langer & Norman's [1]
    FMD expression and Madau & Dickinson's [2] SFRD expression. In a future
    version this will be customisable.

    [1] https://iopscience.iop.org/article/10.1086/500363

    [2] https://arxiv.org/abs/1403.0007

    Arguments:
        df {pandas.dataframe} -- The dataframe of stellar data
                                 (typically the output from
                                 takahe.load.from_file())
        Z {float} -- The metallicity to compute at, expressed as a
                     fraction of solar metallicity.

    Keyword Arguments:
        extra_lt {callable} -- A lambda representing the extra lifetime
                               term you wish to specify. This lambda
                               receives one argument - the dataframe
                               under consideration (so you can specify
```

```
30                                an extra lifetime term that depends on
31                                the stellar properties). If None, then an
32                                empty lambda will be used.
33                                (default: {None})
34
35        Returns:
36            {takahe.histogram} -- A histogram representing the total event
37                                  rate.
38        """
39
40        if extra_lt == None:
41            extra_lt = lambda df: 0
42
43        # Numexpr causes our computations to break.
44        # This is a "feature" of numexpr as far as I can tell
45        pd.set_option('compute.use_numexpr', False)
46
47        lin_edges = np.linspace(0, takahe.constants.HUBBLE_TIME, 51)
48
49        total_event_rate = takahe.histogram.histogram(edges=lin_edges)
50        total_SFRD = takahe.histogram.histogram(edges=lin_edges)
51        Z_prev = 0.0
52
53        events_histograms = dict()
54
55        df.drop(df[df['m1'] > takahe.constants.MASS_CUTOFF].index,
          ↪  inplace=True)
56        df.drop(df[df['m2'] > takahe.constants.MASS_CUTOFF].index,
          ↪  inplace=True)
57
58        G = 6.67430e-11
59        c = 299792458
60
61        # Highly eccentric orbits lead to division by zero.
62        df.drop(df[df['e0'] == 1].index, inplace=True)
63
64        # Unit Conversions:
65        df['a0'] *= (69550 * 1000) # Solar Radius -> Metre
66        df['m1'] *= 1.989e30 # Solar Mass -> Kilogram
```

```
67      df['m2'] *= 1.989e30 # Solar Mass -> Kilogram
68
69      # Introduce some temporary terms, to make computation easier
70      df['beta'] = ((64/5) * G**3 * df['m1'] * df['m2']
71                           * (df['m1'] + df['m2'])
72                           / (c**5))
73
74      df['circ'] = df['a0']**4 / (4*df['beta'])
75
76      df['divisor'] = ((1 - df['e0'] ** (7/4)) ** (1/5)
77                   *  (1+121/304 * df['e0'] ** 2))
78
79      df['coalescence_time'] = ((df['circ'] * (1-df['e0']**2)**(7/2)
80                             / df['divisor'])
81                             / (1e9 * 60 * 60 * 24 * 365.25))
82
83      df['lifetime'] = (df['evolution_age'] / 1e9
84                     +  df['rejuvenation_age'] / 1e9
85                     +  df['coalescence_time']
86                     +  extra_lt(df)
87                     )
88
89      # Unit Conversions (back):
90      df['a0'] /= (69550 * 1000) # Metre -> Solar Radius
91      df['m1'] /= (1.989e30) # Kilogram -> Solar Mass
92      df['m2'] /= (1.989e30) # As above
93
94      # The minimum lifetime of a star is ~3 Myr, so introduce
95      # an artificial cutoff there.
96      df['lifetime'] = np.maximum(df.lifetime, 0.003)
97
98      # I don't think it's required but just in case
99      df.reset_index(drop=True, inplace=True)
100
101     # Remove temporary columns
102     population_at = df.drop(columns=['beta',
103                                      'coalescence_time',
104                                      'evolution_age',
105                                      'rejuvenation_age',
```

```
                                    'circ',
                                    'divisor'
                                   ],
                            inplace=False)

    # population_at['bins'] = pd.cut(population_at.lifetime,
    ↪  np.append([0.0], 10**lin_edges / 1e9), right=False)
    # That's a quick reminder to myself about BPASS binning potentially

    # Perform the binning
    # Thank you to Max Briel for the tip here
    population_at['bins'] = pd.cut(population_at.lifetime,
                                   lin_edges,
                                   right=False)

    bin_widths = lin_edges

    out_df = population_at[["bins", "weight"]].groupby("bins").sum()

    z = takahe.helpers.lookback_to_redshift(bin_widths)

    for Zi in takahe.constants.BPASS_METALLICITIES:
        Zi = takahe.helpers.format_metallicity(Zi)
        Zc = np.mean([Zi, Z_prev])

        if Zi > Z:
            break

        SFRDi = gammainc(0.84, Zc**2 * 10**(0.3*z)) \
                               * 0.015 * (1+z)**2.7 / \
                               (1+((1+z)/2.9)**5.6)

        Z_prev = Zi

        SFRD = takahe.histogram.histogram(edges=z)
        SFRD.fill(z, w=SFRDi)

        SFRD = SFRD - total_SFRD # remove prior SFRD contributions

```

```
144             total_SFRD += SFRD
145
146         DTDi = out_df.values.ravel() / 1e6 / np.diff(bin_widths)
147
148         DTDi = np.append(DTDi, DTDi[-1]) # Because DTDi isn't long enough?
149
150         DTD = takahe.histogram.histogram(edges=lin_edges)
151         DTD.fill(lin_edges, w=DTDi)
152
153         events = takahe.histogram.histogram(edges=lin_edges)
154
155         for i in range(1, len(bin_widths)):
156             t1 = max(bin_widths[i-1], 0.0000001)
157
158             t2 = bin_widths[i]
159
160             this_SFR = SFRD.integral(t1, t2) * 1e9
161
162             this_SFR /= (1e-3)**3
163
164             # Convolve the SFH with the DTD to get the event rates
165             for j in range(i):
166                 t1_prime = t2 - bin_widths[j]
167                 t2_prime = t2 - bin_widths[j+1]
168
169                 events_in_bin = DTD.integral(t2_prime, t1_prime)
170
171                 events.fill(bin_widths[j], events_in_bin * this_SFR)
172
173         # Normalise to years:
174         events /= (np.diff(bin_widths) * 1e9)
175
176         events_histograms[str(Z)] = events
177
178         total_SFRD._values += (SFRD._values)
179         total_event_rate._values += events._values
180
181         return total_event_rate
182
```

```
183 def composite_event_rates(dataframes, extra_lt=None):
184     """Computes the event rate for a variety of stars at different
185     metallicities.
186 
187     Computes an ensemble event rate. Makes the same assumptions that
188     single_event_rate does (because it calls it), namely:
189 
190     - Uses Langer & Norman's [1] FMD expression and Madau & Dickinson's
191     [2] SFRD expression. In a future version this will be customisable.
192 
193     [1] https://iopscience.iop.org/article/10.1086/500363
194 
195     [2] https://arxiv.org/abs/1403.0007
196 
197     Arguments:
198         dataframes {dict} -- A dictionary of dataframes of stellar
199                              properties, indexed by fractional
200                              metallicity (as string).
201 
202     Keyword Arguments:
203         extra_lt {callable} -- A lambda representing the extra lifetime
204                                term you wish to specify. This lambda
205                                receives one argument - the dataframe
206                                under consideration (so you can specify
207                                an extra lifetime term that depends on
208                                the stellar properties). If None, then an
209                                empty lambda will be used.
210                                (default: {None})
211 
212     Returns:
213         {takahe.histogram} -- A histogram representing the total event
214                               rate for your ensemble.
215     """
216 
217     if extra_lt == None:
218         extra_lt = lambda df: 0
219 
220     lin_edges = np.linspace(0, takahe.constants.HUBBLE_TIME, 51)
221 
```

```
    total_event_rate = takahe.histogram.histogram(edges=lin_edges)
    total_SFRD = takahe.histogram.histogram(edges=lin_edges)

    for index in range(len(takahe.constants.BPASS_METALLICITIES)):
        Zi = takahe.constants.BPASS_METALLICITIES[index]
        Z = takahe.helpers.format_metallicity(Zi)

        df = dataframes[str(Z)]

        events = single_event_rate(df, Z, extra_lt)
        total_event_rate += events

    return total_event_rate
```

Appendix C

Takahe Period-Eccentricity Distributions

```
@np.vectorize
def evolve_system(a0, e0, m1, m2):
    """Evolves a binary system until merger or the age of the Universe.

    # TODO: [description]

    Decorators:
        np.vectorize

    Arguments:
        a0 {float} -- The initial semimajor axis in Solar Radii
        e0 {float} -- The initial eccentricity
        m1 {float} -- The mass of the primary in Solar Masses
        m2 {float} -- The mass of the secondary in Solar Masses

    Returns:
        {tuple} -- The initial and final parameters:
                    - The initial eccentricity
                    - The initial period (days)
                    - The final eccentricity
                    - The final period (days)
                    - The number of samples
    """
    global pbar, matrix_elements

    params = [m1, m2]

    a, e = takahe.helpers.integrate(a0, e0, params)
```

```
29
30      af = a[-1]
31
32      e0 = e0
33      ef = e[-1]
34      P = takahe.helpers.compute_period(a, m1, m2)
35      P0, Pf = P[0], P[-1]
36
37      nbin_width = 1e-2 # Todo: customisable
38      mbin_width = 1e-2
39
40      logP = np.log10(P)
41
42      nbins = int(1 // nbin_width)
43      mbins = int(8 // mbin_width)
44
45      bins = max(nbins, mbins)
46
47      binx = np.linspace(-2, 6, bins)
48      biny = np.linspace(0,  1, bins)
49
50      ret = stats.binned_statistic_2d(logP, e, logP, 'count', bins=[binx,
        ↪  biny])
51
52      bin_counts = ret.statistic
53
54      pbar.update(1)
55
56      matrix_elements.append(bin_counts)
57
58  def period_eccentricity(in_df, transient_type="NSNS"):
59      """Computes the period-eccentricity distribution for an ensemble.
60
61      Evolves the ensemble from 0 to the Hubble time to see how systems
62      behave.
63
64      Strictly this is an SMA-eccentricity routine but the period is
65      computable from the SMA using Kepler's third law.
66
```

```
67      Adds FOUR new columns to the DataFrame:
68          - a (the array of SMAs for this star)
69          - e (the array of eccentricities for this star)
70          - af (the final SMA for this star)
71          - ef (the final eccentricity for this star)
72
73      # Todo: Correct implementation.
74      # Todo: Return a period array.
75
76      Arguments:
77          in_df {pd.DataFrame} -- The DataFrame representing your ensemble.
78
79      Keyword Arguments:
80          transient_type {str} -- The transient type (NSNS, NSBH, BHBH)
81                                  under consideration. (default: {"NSNS"})
82
83      Returns:
84          {pd.DataFrame} -- The DataFrame with the a and e arrays added
85                            as new columns.
86      """
87
88      global pbar, matrix_elements
89
90      # Don't know why we have to do this, computation breaks otherwise.
91      pd.set_option('compute.use_numexpr', False)
92
93      # Now we mask out what we're not interested in.
94      df = takahe.event_rates.filter_transients(in_df, transient_type)
95
96      # Highly eccentric orbits lead to division by zero.
97      df.drop(df[df['e0'] == 1].index, inplace=True)
98
99      if len(df) == 0:
100         return np.zeros((800-2, 800-2))
101
102     matrix_elements = []
103     with tqdm(total=len(df)) as pbar:
104         evolve_system(df['a0'].values,
105                       df['e0'].values,
```

```
106                         df['m1'].values,
107                         df['m2'].values)
108
109
110     C = np.sum(matrix_elements, 0)
111
112     return C
```

Appendix D

Julia Integration Routine

```
function Adjust(da_or_de, beta, alpha)
    return da_or_de^beta + alpha
end

function integrate(a0, e0, p)
    """
    General purpose integrator for Nyadzani & Razzaque eqns for a & e.

    Params:
        a0 - The initial semimajor axis, measured in solar radii
        e0 - The initial eccentricity, dimensionless
        p  - A vector of parameters:
                 p[1] = m1 (units: Solar Mass)
                 p[2] = m2 (units: Solar Mass)
                 p[3] = Beta for LT Adjustment (units: dimensionless)
                 p[4] = Alpha for LT adjustment (units: [unit]/s)
                 p[5] = Lifetime (evolution + rejuvenation)

    Returns:
        A  - An array of the semimajor axes of the binary system over time.
    ↪  (Solar Mass)
        E  - An array of the eccentricities of the binary system over time.
    ↪  (no dim.)
    """

    Solar_Mass   = 1.989e30  # kg
    Solar_Radius = 696340000.0 # m
               G = 6.67e-11  # m^3 kg^-1 s^-2
               c = 299792458.0 # m/s

               A = Float64[]
```

```
30                  E = Float64[]
31                  H = Float64[]
32
33      m1, m2, expo, alph, evotime = p[1], p[2], p[3], p[4], p[5]
34
35      # number of seconds in a year
36      seconds_per_year = 60 * 60 * 24 * 365.25
37
38      ########################
39      ##     Unit Check     #
40      ########################
41      a = a0 * Solar_Radius # Meters
42      e = e0                # Dimensionless
43      m1 = m1 * Solar_Mass  # Kilogram
44      m2 = m2 * Solar_Mass  # Kilogram
45      ########################
46      ##   End Unit Check   #
47      ########################
48
49      # Beta has units m^4 / s
50      beta = ((64/5) * G^3 * m1 * m2 * (m1 + m2) / (c^5))
51
52      A = push!(A, a)
53      E = push!(E, e)
54      H = push!(H, 0.0)
55
56      total_time = 0
57
58      # Integrate until past the end of the universe, or a 10km orbit
59      while total_time/seconds_per_year + evotime < 1e11 && a > 1e4
60          initial_da = (- beta / ( (a^3) * (1 - e^2)^(7/2) ))
61          da = initial_da * (1 + (73/24) * e^2 + (37/96) * e^4)
62
63          intial_de = (((-19/12) * beta) / (a^4*(1-e^2)^(5/2)))
64          de = intial_de * (e + (121/304) * e^3) # Units: s^-1
65
66          # da = Adjust(da, expo, alph)
67          # de = Adjust(de, expo, alph)
68
```

```
69          timeA = abs(1e-2 * a/da)
70          if e > 1e-10
71              timeE = abs(1e-2 * e/de)
72          else
73              de = 0
74              e = 1e-10
75              timeE = timeA * 10
76          end
77
78          dt2 = (evotime +
            ↪  total_time/seconds_per_year)*0.23076752*0.5*seconds_per_year
79
80          dt = min(timeE, timeA, dt2)
81
82          a = a + dt * da
83          e = e + dt * de
84
85          A = push!(A, a)
86          E = push!(E, e)
87          H = push!(H, dt)
88
89          total_time = total_time + dt
90      end
91
92      # Solar Radii, Dimensionless
93      return A / Solar_Radius, E, H
94  end
```

Appendix E

Highly Eccentric Mergers Code

```
import copy
from os import path

import numpy as np
import pandas as pd
import takahe
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from matplotlib import cm
from tqdm import tqdm

metallicity = "020"
kick        = "Bray"
A           = np.arange(0, 310, 10)
B           = np.arange(-200, 110, 10)
dt          = 1e6
ecc_cutoff  = 0.8

def compute(m1, m2, e0, a0, age, pbar=None):
    """Computational workhorse function.

    Evolves a DNS system to either its coalesence or 1e11 years. Returns
    relevant parameters for this script.

    Arguments:
      m1 {float} -- Mass of the primary star
      m2 {float} -- Mass of the secondary star
      e0 {float} -- Initial eccentricity
```

```
        a0 {float} -- Initial SMA
        age {float} -- The age (evolution + rejuvenation)

    Keyword Arguments:
        pbar {tqdm} -- A progress bar to update. (default: {None})

    Returns:
        tuple -- The required parameters:
                    - Final Eccentricity
                    - Coalescence time (Gyr)
                    - Final SMA
                    - Eccentricity at the Hubble Time
    """
    a, e, h = takahe.evolve.evolve_system(a0, e0, m1, m2,
                                          only_arrays=True,
                                          return_value=True,
                                          evotime=age)

    sec_gyr = takahe.constants.SECONDS_PER_GYR
    tH = takahe.constants.HUBBLE_TIME

    i = np.argmin(np.abs(h/sec_gyr - tH))

    if pbar is not None:
        pbar.update(1)

    return e[-1], np.sum(h) / sec_gyr, a[-1], e[i]
# Vectorise the function (but ignore the pbar since that isn't an array)
compute = np.vectorize(compute, excluded=['pbar'])

with tqdm(total=31*31) as pbar:
    for i in range(31):
        for j in range(31):
            ecc_dist = takahe.histogram.histogram(0, 1, 20)
            alpha, beta = A[i], B[j]

            if kick == "Bray":
                fname = "" # omitted for brevity
                df = pd.read_hdf(fname, f'/{beta}')
```

```
68              else:
69                  fname = "" # omitted for brevity
70                  df = takahe.load.from_file(fname)
71
72              df = takahe.event_rates.filter_transients(df, 'NSNS')
73
74              df['age'] = df['evolution_age'] + df['rejuvenation_age']
75
76              # Reindented for clarity
77              df['ef'], df['ct'], df['af'], df['e_at_th'] = (
78                                  compute(df['m1'].values,
79                                          df['m2'].values,
80                                          df['e0'].values,
81                                          df['a0'].values,
82                                          df['age'].values
83                                          )
84                                                          )
85
86              df['coalescence_time'] = (df['age'] / 1e9 + df['ct'])
87
88              df['P_final'] = takahe.helpers.compute_period(df['m1'].values,
89                                                            df['m2'].values,
90                                                            df['af'].values)
91
92              df['P_final'] = df['P_final'] * (24 * 60 * 60)
93
94              df['gw_freq'] = 2 / df['P_final']
95
96              # Filter out uninteresting systems
97              df = df[df['ef'] > ecc_cutoff]
98              df = df[df['coalescence_time'] > takahe.constants.HUBBLE_TIME]
99
100             # So df contains all systems which have e > 0.8 and
101             # a merger time that is > the age of the universe.
102             # now we say what is the eccentricity distribution for today
103
104             ecc_dist.fill(df['e_at_th'].values)
105
106             ecc_dist.to_pickle(f'data/dists/ecc_dist_{alpha}_{beta}.p')
```

```
107
108                 pbar.update(1)
109             # end alpha loop
110         # end beta loop
```

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Isa Isa vulagi lasa dina
Nomu lako au na rarawa kina
Cava beka ko a mai cakava,
Nomu lako au na sega ni lasa.

Alas, alas, most welcome guest,
your going fills me with sorrow.
Whatever the reason you came,
I feel bereft at your leaving.

The Fijian Farewell Song (accompanying rough English translation)